Humanoid Locomotion and Manipulation: Current Progress and Challenges in Control, Planning, and Learning

Zhaoyuan Gu, *Student Member, IEEE*, Junheng Li, *Student Member, IEEE*, Wenlan Shen, *Graduate Student Member, IEEE*, Wenhao Yu, Zhaoming Xie, *Member, IEEE*, Stephen McCrory, Xianyi Cheng, *Member, IEEE*, Abdulaziz Shamsah, *Member, IEEE*, Robert Griffin, *Member, IEEE*, C. Karen Liu, Abderrahmane Kheddar, *Fellow, IEEE*, Xue Bin Peng, Yuke Zhu, *Senior Member, IEEE*, Guanya Shi, *Member, IEEE*, Quan Nguyen, *Member, IEEE*, Gordon Cheng, *Fellow, IEEE*, Huijun Gao, *Fellow, IEEE*, and Ye Zhao, *Senior Member, IEEE*

***Abstract*—Humanoid robots hold great potential to perform various human-level skills, involving unified locomotion and manipulation in real-world settings. Driven by advances in machine learning and the strength of existing model-based approaches, these capabilities have progressed rapidly, but often separately. This survey offers a comprehensive overview of the state-of-the-art in humanoid locomotion and manipulation, with a focus on control, planning, and learning methods. We first review the model-based methods that have been the backbone of humanoid robotics for the past three decades. We discuss contact planning, motion planning, and whole-body control, highlighting the tradeoffs between model fidelity and computational efficiency. Then, the focus is shifted to examine emerging learning-based methods, with an emphasis on reinforcement and imitation learning that enhance the robustness and versatility of loco-manipulation skills. Furthermore, we assess the potential of integrating foundation models with humanoid embodiments to enable the development of generalist humanoid agents. This survey also highlights the emerging role of tactile sensing, particularly whole-body tactile feedback, as a crucial modality for handling contact-rich interactions. Finally, we compare the strengths and limitations of model-based and learning-based paradigms from multiple perspectives, such as robustness, computational efficiency, versatility, and generalizability, and suggest potential solutions to existing challenges.**

***Index Terms*—Foundation models (FMs), humanoid robotics, imitation learning (IL), loco-manipulation, model predictive control, whole-body control, whole-body tactile sensing.**

Received 3 January 2025; revised 11 April 2025; accepted 12 May 2025. Date of publication 26 March 2026; date of current version 20 April 2026. Recommended by Technical Editor C. Yang and Senior Editor X. Chen. This work was supported in part by the Office of Naval Research (ONR) under Grant N000142312223, in part by the National Science Foundation (NSF) under Grant IIS-1924978, Grant CMMI-2144309, and Grant FRR-2328254, and in part by the United States Department of Agriculture (USDA) under Grant 2023-67021-41397. *(Corresponding authors: Ye Zhao; Huijun Gao.)*

Please see the Acknowledgment section of this article for the author affiliations.

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TMECH.2025.3579247.

Digital Object Identifier 10.1109/TMECH.2025.3579247

I. Introduction

A HUMANOID robot refers to any anthropomorphic robot that resembles the form of a human [7]. Typically, a humanoid robot possesses a torso, two arms, and two legs, though the degree of anthropomorphism may vary. For instance, some humanoid robots feature simple hands or wheeled legs [8]. The primary focus of this review is on humanoid robots that replicate human morphologies and functionalities, rather than those that closely mimic human-like visual appearance and external aesthetics. Specifically, this survey examines bipedal humanoid robots, with an emphasis on the whole-body motion of humanoid robots instead of on the dexterous manipulation of multifinger hands.

Humanoid robots are well suited for executing human-level tasks, as they are built to (ideally) replicate human motions and achieve various whole-body loco-manipulation tasks, e.g., applications ranging from manufacturing to services, as shown in Fig. 1. Their anthropomorphism makes them stand out from other robot forms in performing human-like tasks, such as payload carrying over stairs, especially in environments designed for humans. Furthermore, humanoid robots can interact with humans for physical collaboration tasks, such as collaboratively moving a large table upstairs with human assistance. However, simultaneously achieving these intricate tasks while addressing highly complex robot dynamics is still challenging, let alone safe physical collaborations with humans and/or operations in unstructured environments. As a promising direction to solve this problem, humanoids can exploit the abundance of data collected from humans to quickly acquire motor and cognitive skills due to their human-like morphology. Therefore, scaling human data and computational resources for humanoid embodiment is potentially a fast route to embodied intelligence.

Embodied intelligence at large is thriving at an unprecedented pace. Perception algorithms can detect, classify, and segment a wide variety of objects in real time. Model-based methods that

1083-4435 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

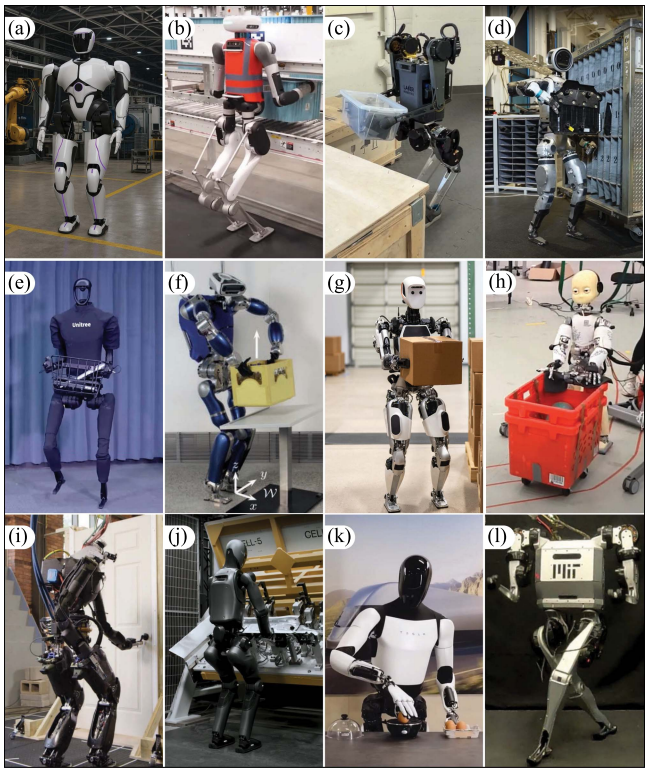


Fig. 1. Humanoids executing locomotion and manipulation tasks. (a) Q-2 humanoid working in a factory [1]. (b)–(g) Object pick and place by Digit, Hector [2], Atlas, H1, TORO [3], and Apollo. (h) IIT Cognitive Universal Body (iCub) pushes a cart [4]. (i) Nadia opens a door [5]. (j)–(k) Object manipulation by Figure 02 and Optimus. (l) MIT humanoid whole-body push recovery [6].

leverage both predictive and reactive control have enabled agile and reliable locomotion and manipulation. Meanwhile, deep learning policies have demonstrated convincing control results on robot hardware through exploration and imitation. Large foundation models (FMs) trained on massive, Internet-scale datasets are beginning to show cognitive capabilities of open-world reasoning. Consequently, the building of autonomous humanoid robots for real-world applications has become possible, leading to the emergence of many humanoid robot companies and concrete deployment applications.

Acknowledging the rapid advancements in humanoid robotics, this article reviews recent developments in humanoid locomotion and manipulation (HLM). As laid out in Fig. 2, humanoid robotics is a multidisciplinary field that spans domains in design, actuation, sensing, control, planning, and decision-making. In this survey, we mainly examine task planning, motion planning, policy learning, and control from the perspective of model-based and learning-based methods. We first synthesize traditional model-based methods for planning and control. Then, we shift our focus to more recent learning-based approaches, especially those leveraging reinforcement learning (RL), imitation learning (IL), and FMs. Model-based methods serve as the cornerstone for enabling HLM capabilities. Over the past decade, planning and control techniques have shown a trend of converging to the predictive-reactive control hierarchy, employing a whole-body model predictive control (MPC) or simplified model [centroidal dynamics (CD)] MPC coupled with local task-space whole-body control (WBC) [9]. These planning and control techniques are usually formulated as optimal control problems (OCPs) that are solved by off-the-shelf or customized numerical solvers. Although these numerical optimization methods are well-established, research continues to focus on enhancing their computational efficiency, numerical stability, robustness, and scalability for high-dimensional systems.

Learning-based approaches have witnessed a rapid surge in humanoid robotics and achieved impressive results that attract an increasing number of researchers to the field. Among the diverse learning approaches, RL has proven its ability to achieve robust motor skills. However, despite its ability to discover novel behaviors via trial and error, pure RL without demonstration data is often prohibitively inefficient for HLM tasks, which are characterized by high degrees-of-freedom robots and sparse reward settings. Therefore, sim-to-real RL, where training is conducted in simulation before transferring to the physical robot, has become a prevalent method, though it faces the challenge of bridging sim-to-real gaps. From the existing literature, we observe that sim-to-real RL has gained significant traction for achieving humanoid whole-body motion. There is an ongoing debate between model-based and learning-based control, and therefore, we highlight two key points: 1) sim-to-real RL that relies on the accurate dynamics models in a simulator can also be viewed as a "model-based," method, even though the RL algorithm itself does not explicitly model the dynamics; and 2) model-based control and sim-to-real RL do not conflict; on the contrary, they often complement each other and can be combined to achieve a better performance than either method alone.

IL from expert demonstrations has proven to be an efficient method of acquiring motor skills. In pursuit of versatile and generalizable policies through IL, many researchers and companies have focused on scaling data. While robot experience data can be diverse and high-quality, its acquisition is both expensive and time-consuming. Thus, learning from human data, which is abundant and readily available from Internet videos and public datasets, emerges as a pivotal strategy for humanoid robotics. Even though humanoid robots may attain human-level motor skills, a deeper question of embodied intelligence persists: how to learn the intentions (source) behind human actions rather than merely replicating the observed motions (outcome). It is hypothesized that understanding human intention is achieved via FMs that are capable of semantic interpretation of the environment and the task. This hypothesis motivates us to include FMs as a part of our survey.

The remarkable success of FMs has sparked a surge in *general* robotics research, as highlighted by several comprehensive surveys [10], [11]. This article surveys the application of FMs specifically for humanoid robots. FMs offer a promising solution to the persistent challenge of generalizability in robotics by efficiently harnessing internet-scale datasets to acquire extensive knowledge. Training end-to-end FMs for humanoid applications is an emerging area, poised for significant future advancements.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

Fig. 2. Survey begins by defining relevant concepts of humanoid robots and their locomotion and manipulation capabilities. Centered around achieving humanoid loco-manipulation tasks, the core of this survey delves into two main categories of methods: the traditional planning and control approaches, such as contact planning, motion planning, and control, as well as the emerging learning-based approaches, including skill learning and FMs. In addition, this survey highlights whole-body tactile sensing as a crucial modality to achieve contact-rich loco-manipulation.

A. Survey Roadmap

For the selection criteria of papers cited in this survey, we prioritize topics most relevant to humanoid loco-manipulation capabilities, including tactile sensing [12], model-based planning [13], and control [9], as well as learning-based methods such as RL [14], [15], IL [16], [17], and FMs [10], [11]. Distinct from survey papers on these topics, our article aims to present a broad overview with a focus on humanoid loco-manipulation.

Each of these topics has extensive studies. For each topic, we aim to spotlight the following three categories of papers:

1) foundational studies that established key principles in these areas;
2) recent works representing the state of the art;
3) survey papers for further reading.

Our survey spans 30 years of research. Specifically, model-based planning and control papers have been collected since 1990, and learning-based methods predominantly span the past 20 years. Compared with existing survey reviews on humanoid robotics [7], [18], this survey collects up-to-date papers and introduces in detail the new paradigm of learning-based methods.

As shown in Fig. 2, this survey is organized in the following order. We first establish the background, defining humanoid robots and the key capability of locomotion and manipulation in Section II. We detail whole-body tactile sensing in Section III. We then present traditional approaches including contact planning (see Section IV), motion planning (see Section V), and control (see Section VI). We then examine learning-based algorithms. In Section VII, we explore approaches using RL and IL to acquire loco-manipulation skills. In Section VIII, we discuss how FMs have become the backbone of semantic understanding and decision-making for effective humanoid task planning. Finally, Section IX concludes this article.

B. Survey Contributions

We summarize our core contributions as follows:

1) The survey serves as an effective resource for graduate students and researchers new to the field, offering a comprehensive review of humanoid technical methods, while also providing perspectives for humanoid experts in academia and industry with the latest advancements.
2) We advance the understanding of loco-manipulation in bipedal humanoid robots by clearly defining the problem, surveying state-of-the-art methods and capabilities,

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

and outlining remaining challenges along with promising solutions.

3) We provide a clear comparison between model-based and learning-based methods, and their corresponding loco-manipulation skills achieved on humanoid robots. This comparison offers valuable insights for researchers deciding their methods.

II. Background

A. Bipedal Locomotion

Bipedal locomotion is a significant characteristic of humanoid robots. Therefore, in the past three decades, bipedal locomotion has been a prolific field of humanoid research. Interested readers can refer to the excellent reviews (most of which are recent) [19], [20], [21], and the monographs [22], [23]. In summary, model-based bipedal locomotion has evolved significantly, progressing from passive walking [24], [25] to quasi-static walking [26], and then to dynamic walking [22].

Bipedal locomotion under external perturbations and force loads has been extensively studied. Such capabilities lay the foundation for simultaneous locomotion and manipulation, the focus of this survey. Model-based methods are among the early works that demonstrate these capabilities [2], [27], [28]. In addition to model-based methods, bipedal locomotion has also been successfully addressed by learning-based methods [29], [30], [31], particularly in the context of periodic motions on flat surfaces. Furthermore, learning-based approaches have also demonstrated capabilities in more complex settings, such as running [32], jumping [33], and handling nonperiodic motions such as stair climbing [34] and parkour [35]. Similar to the trend in model-based methods, learning-based methods have further extended their capabilities to handle external forces and payloads [36], [37].

Researchers have explored energy efficiency in bipedal locomotion by examining the cost of transport (COT), which designates energy expenditure per unit distance traveled, normalized by body weight. Early work in passive dynamic walking [24], [25] achieved higher efficiency than humans but offered limited versatility. Meanwhile, today's humanoid robots $(\mathrm{COT} > 0.7)$ are significantly less efficient than humans $(\mathrm{COT} = 0.2)$ [38]. Therefore, efficient walking continues to motivate hardware designs that incorporate passive-compliant components for energy storage [39], [40] and control algorithms that exploit the system's natural dynamics [41].

In the event of a locomotion failure, the ability to withstand and recover from a fall is essential for humanoid robots. Recent efforts have made impressive strides toward this goal, particularly in key safety measures for these inherently unstable systems, including fall detection [42], fall reaction [43]. Notably, Zhang et al. [42] presented a compelling approach to fall detection by identifying and analyzing critical robot states, offering valuable insights into prefall dynamics. Their work stands out for not only its analytical depth but also its practical relevance—culminating in a novel adjustment strategy within the control framework.

B. Whole-Body Manipulation

The idea of whole-body manipulation [44] was originally studied in the whole-arm manipulation community [45]. Whole arm manipulators were designed and built to explore the benefit of manipulating objects with all surfaces of a robot manipulator [46]. This brings a unique challenge that manifests itself in all the system levels in perception, estimation, planning, and control. Numerous breakthroughs in mechanical design, control, and planning have been achieved in the endeavor to address the challenges of whole-body manipulation.

On the mechanical design side, robots made with soft materials and full-body sensing, such as Punyo [47], provide whole-body manipulation capability in a built-in manner.

For control, the coordinative and contact-rich nature requires forceful and compliant control. Traditionally, robot arms were hard-coded to switch across different control strategies according to task requirements [48]. However, it is still unclear how to define and enumerate the control strategies for whole-body manipulation. In addition, a general control framework that can take in the sensor data, perform state estimation, and reactively control each body contact has yet to exist.

From the planning perspective, the challenge of whole-body manipulation can be potentially alleviated via human behavior imitation algorithms [49], [50]. Most of these works focus on simple manipulation strategies such as whole-body grasping and pushing. To enable the robot to mimic more complex human whole-body manipulation behaviors, it is important to address the cross-morphology gap between humans and humanoids.

To achieve humanoid whole-body manipulation, full-stack system integration at all system levels is crucial. Such general systems require innovations in advanced hardware and algorithm architecture, including whole-body sensing [51], robot designs with compliance, and force control capabilities for reactive manipulation.

C. Loco-Manipulation

One of the key features of humanoid robots is their ability to simultaneously perform locomotion and manipulation (abbreviated as loco-manipulation hereafter) tasks. As suggested by its name, loco-manipulation involves both the movement of objects through manipulation and the mobility of the robot itself through locomotion. In a more general case of *whole-body* loco-manipulation, the *whole-body* refers to the use of all body surfaces to interact with the environment. We summarize the relationship between loco-manipulation, and whole-body manipulation in Table I. As shown in Fig. 3, loco-manipulation considers the movement of the robot itself while it manipulates an object, whereas whole-body manipulation emphasizes leveraging all accessible robot contact surfaces, such as using the chest as extra support to move large objects.

Loco-manipulation for humanoid robots is particularly challenging. Humanoid robots have a smaller support region on the ground and a higher center of mass, which is challenging for dynamic balance. Therefore, early humanoid frameworks focus on separate control for locomotion and manipulation. For example, in locomotion tasks, most studies constrain the upper body to

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TAXONOMY OF WHOLE-BODY LOCOMOTION AND MANIPULATION

	(a) Whole-body Manipulation	(b) Whole-body Loco-manipulation	(c) Loco-manipulation
Object movement (Manipulation)	✓	✓	✓
Robot self-mobility (Locomotion)	✗	✓	✓
All surface interaction (Whole-body)	✓	✓	✗

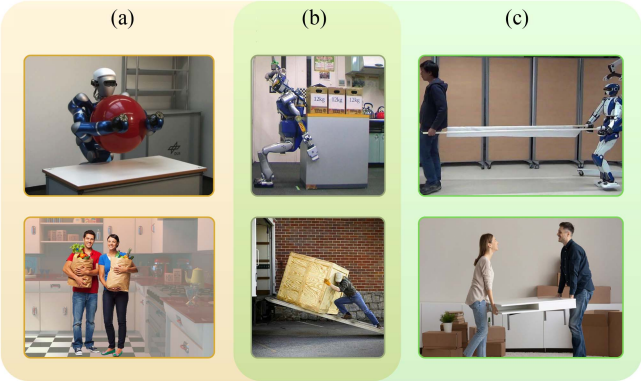


Fig. 3. (a) Whole-body manipulation exemplified by human and humanoid Justin [52] interacting with objects using all surfaces. (c) Loco-manipulation involves simultaneous locomotion and manipulation, as shown in the collaborative tasks performed by humans and a humanoid [28]. (b) Whole-body loco-manipulation is an intersection of (a) and (c), as exemplified by a human and a humanoid HRP-4 [53] pushing heavy objects using their legs and arms. (a) Whole-body manipulation. (b) Whole-body loco-manipulation. (c) Loco-manipulation.

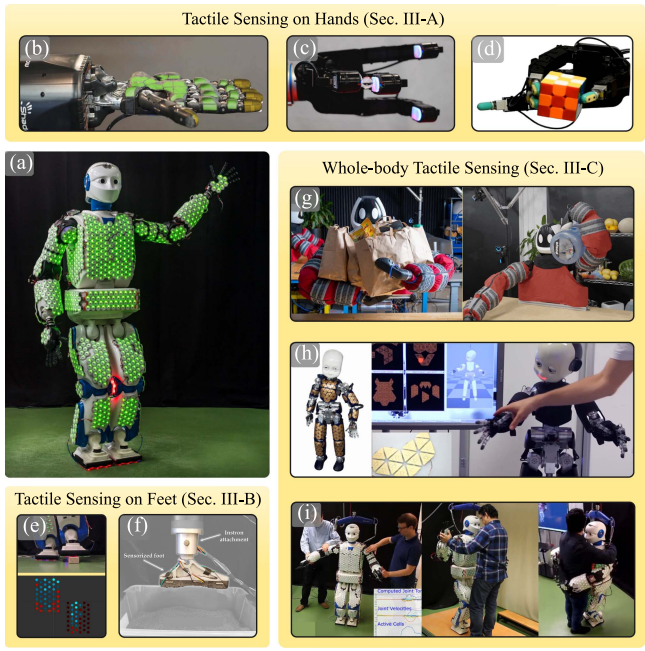


Fig. 4. Tactile sensing on humanoid robots, exemplified by (a) REEM-C fully covered with artificial skin [51] (image copyright: A. Eckert), which cover three body regions: hand, feet, and the whole body. (i) Hand tactile sensors demonstrated by (b) Shadow-Dexterous-Hand equipped with tactile sensors on palm and fingertips [70]. (c) Allegro Hand equipped with DIGIT sensors [68], and (d) BioTac [82] tactile sensors for dexterous manipulation; (ii) Tactile sensors on foot soles for (e) obstacle recognition [79] and (f) terrain classification [80]; (iii) Whole-body tactile sensors for (g) whole-body manipulation by Punyo-1 [47] and whole-body human–robot interaction by (h) iCub [83] and (i) REEM-C [84], [85].

remain upright, which simplifies the whole-body problem to a bipedal locomotion problem that considers only the lower limbs. Conversely, in most table-top manipulation tasks, the lower body of the humanoid remains stationary [54], [55]. In such cases, any external force exerted on the upper body is treated as a disturbance to the legs, whose goal is to solely maintain balance. On the contrary, there is no such categorization of contacts in loco-manipulation: all contacts contribute simultaneously to achieve the task and balance.

III. TACTILE SENSING

HLM involve extensive physical interactions with the environment and objects, requiring multimodal sensing for understanding the environment, tracking manipulated objects, and evaluating how contact impacts the balance of both the robot and the objects. Visual sensors have shown effectiveness in object tracking and simultaneous localization and mapping [56], while proprioceptive sensors are usually combined to estimate contact information in contact-rich tasks [57]. These sensory modalities have been widely adopted in existing systems and have been thoroughly reviewed in the literature [18]. In contrast, tactile sensing, despite being uniquely suited for capturing detailed contact information critical for both locomotion and manipulation, has received relatively limited attention in the existing humanoid literature. This survey specifically highlights tactile sensing to address this gap and to underscore its complementary role alongside other sensing modalities in humanoid systems.

Numerous studies have developed tactile sensors based on various transduction principles that can sense normal and tangential forces, vibration, temperature, and precontact proximity information. Comparative studies of various sensor designs can be found in [12], [58], and [59]. This survey instead focuses on their application in humanoid loco-manipulation, categorized into the following three areas:

1) tactile sensing on hands;
2) tactile sensing on foot soles;
3) whole-body tactile sensing.

The following sections review recent advancements in each domain, emphasizing their roles in balancing control, scheduling contacts, and enhancing interaction capabilities, as illustrated in Fig. 4 and Table II.

A. Tactile Sensing on Hands

Tactile sensors on dexterous hands provide contact information, addressing the challenges in object manipulation such as grasped object controllability and object property estimation. In this section, we survey studies that integrate tactile perception into the control, planning, and learning of complex manipulation tasks. Due to the similar nature of contact-rich interactions, the tactile sensing techniques on hands also offer valuable insights

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ROLES AND CHALLENGES OF TACTILE SENSING IN LOCOMOTION AND MANIPULATION APPLICATIONS

Application	Roles of tactile sensing	Body region involved	Challenges	Solutions
Manipulation	• Grasping force and stability regulation [60, 61, 62, 63] • Object property and state estimation [64, 65, 66] • Contact dynamics estimation [67, 68] • Input modality for RL/IL/FM training [69, 70, 71, 72, 73]	• Hand (in IHM*) • Whole-body (in WHM*)	• High-dimensional contact space • Tactile physics simulation • Sim-to-real transfer	• Dimensionality reduction [70, 74, 75] • Tactile simulators [76, 77, 78]
Locomotion	• GRF and support region estimation [79] • Terrain property recognition [80, 81]	• Feet	• Sensor durability against heavy weight and various terrains	• Durable materials and mechanical design [79, 80, 81]
Loco-manipulation	• All of above • Unified locomotion and manipulation planning	• Whole-body	• All of Above • Real-time contact reasoning and adaptation	• To be explored

IHM—In-hand manipulation; WBM—Whole-body manipulation.

for whole-body tactile sensing and manipulation, which is discussed in Section III-C.

To achieve the grasping objectives, sensed contact forces serve as real-time feedback in force or impedance control loops to regulate the desired object behavior [60]. Moreover, slip detection and prediction based on tactile sensor data are used to adapt grasping forces, thereby enhancing grasp stability [61], [62], [86].

More complex in-hand manipulation tasks demand interactive perception beyond static object models. Dynamic contact information, including real-time tracking of object states, monitoring contact stability [65], and predicting interaction outcomes [67], [68], i.e., how contact forces affect the balance of both objects and the robot, are crucial to achieving complex interactive behaviors. However, due to the inherent complexity of multicontact dynamics and increased dimensionality of the contact state space, model-based methods still struggle to match human-level dexterity and versatility in multifinger manipulation.

Alternatively, model-free RL has shown the ability to address complex contact interactions. These approaches integrate tactile measurements directly into the state space to train end-to-end policies [69], [70].

B. Tactile Sensing on Feet

Other than manipulation, tactile sensing has started to gain traction for locomotion problems. For legged locomotion, estimation of ground reaction forces (GRFs) and terrain properties is critical for maintaining whole-body stability on diverse, uneven surfaces. While vision and proprioception sensors can provide an indirect estimation of the terrain, these sensing modules lack the capability of accurately estimating GRFs and terrain properties. Tactile sensing on foot soles has the potential to provide direct, unobstructed, and accurate contact measurements, but remains largely underexplored.

To date, tactile sensors for legged systems have been primarily developed for monopods, quadrupeds, and hexapods. Only a few studies have built tactile feet sensors for humanoid robots, with applications such as terrain classification [80] and ground slope recognition [81]. The sensed tactile information should aid the humanoid control and enhance the locomotion performance. A notable work in this direction [79] reconstructs the pressure shape of the foothold, enabling the recognition of uneven terrain and footstep replanning in real time.

Building tactile sensors for humanoid feet is more challenging due to larger impulse and shear force during intermittent ground contact caused by fewer legs and heavier robot weight. Another challenge lies in developing reliable sensors capable of withstanding various terrains, prompting researchers to seek durable materials and dependable mechanical designs.

C. Whole-Body Tactile Sensing

Whole-body tactile sensing extends the aforementioned single-body sensing to all parts of the body, enabling humanoid robots to interact with unknown environments not only by the fingertips or foot soles but also by the arms, legs, and torso.

With explicit tactile feedback, humanoid robots such as iCub and REEM-C have achieved whole-body compliance [84], [87], controlling the contact force from whole-body regions. This level of contact awareness facilitates safe and intuitive physical human–robot interactions including dancing with human [85]. Contact awareness is also useful for improving balance and collision avoidance in unstructured environments.

Large-area tactile sensing significantly enhances a robot's ability to handle *large* objects, including object identification through tactile exploration and whole-body manipulation. For example, Kaboli and Cheng [66] enabled a humanoid robot Nao, covered with artificial skin over its entire upper body, to classify large, heavy objects with different weights and textures. Jain et al. [88] demonstrated whole-arm tactile sensing by reaching objects in cluttered spaces while regulating contact forces across its arms. Close-proximity whole-body capacitive sensing is implemented in [89], enabling a cobotic humanoid with workers close-presence awareness. In addition, with tactile sensors covering their arms and chest, humanoid robots HRP-2 and Punyo-1 can use their entire upper body to grasp and lift large, heavy boxes [63], [90], or various household items [47].

D. Challenges in Tactile Sensing

Current works of tactile-based manipulation, whether in-hand or whole-body, are still largely limited to grasping or simple motions like rolling, pushing, or pick-and-place. Although tactile sensing holds great promise, its integration into more

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

dexterous manipulation or loco-manipulation tasks that involve more dynamic interactions and contact shifting over the full body continues to pose substantial challenges. These include understanding the complex multicontact dynamics, handling the high dimensionality of the sensor data, and addressing the sim-to-real gap.

Several approaches have been explored to address these difficulties. For instance, dimensionality reduction techniques such as spectral clustering, principal component analysis [74], and autoencoders [75] have been employed to manage the high-dimensional input space. Others have proposed simplifying sensor outputs by using boolean contact states (i.e., contact or no-contact) instead of continuous force values in RL training [70]. Furthermore, advances in tactile simulation [76], [77] are making simulated tactile data more accessible, thereby supporting zero-shot sim-to-real transfer [78].

One of the remaining fundamental challenges is to dynamically reason about contacts. This involves not only estimating contact states and static object models but, more crucially, understanding how these contacts and changes of contacts impact the system in real time, including the balance of both the robot and the object. Such information is vital for a planner to make informed decisions and, in a learning framework, can enhance sample efficiency. Addressing this challenge calls for deeper collaboration between the sensing and control communities to determine what tactile information should be abstracted and how to integrate real-time tactile perception into control and planning for humanoid tasks.

Key Takeaways: Tactile sensing is yet an underexplored modality for advancing humanoid loco-manipulation, providing direct contact information necessary for tasks involving complex interactions with environments and objects. While it has improved task performance, achieving human-level dexterity and versatility requires advances in dynamic perception, and multimodal sensing integration to enable systematic, real-time decision-making during interactions. Future directions include optimizing whole-body contact scheduling based on object properties and understanding how contact dynamics affect robot and object balance during loco-manipulation. Moreover, the design of whole-body tactile systems should account for varying sensor resolutions and load requirements, i.e. hands need higher resolution for delicate tasks, while body skin can operate at lower resolution but withstand higher payloads. For further reading, we recommend a survey paper on humanoid tactile sensing [12], and a book chapter on tactile sensing technologies with an emphasis on deployment on humanoid robot [91].

IV. Multicontact Planning (MCP) for Humanoids

MCP remains one of the most challenging tasks in robotics. Specifically, in the context of humanoid whole-body loco-manipulation, a planner ought to solve trajectories that handle rich and intricate interactions with environments or objects. Particularly, beyond robot state trajectories, the planner is also expected to decide *contact position* (or *contact location*), *contact mode*, and *contact force* in a loco-manipulation task. This raises an open problem: how to solve the locomotion and manipulation contact planning problem simultaneously with the whole-body

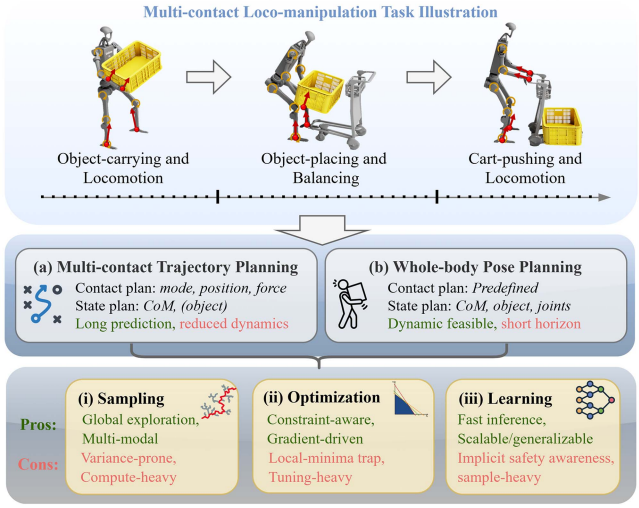


Fig. 5. Illustration of a task sequence for loco-manipulation planning in humanoid robots, involving carrying and placing a box and pushing a cart. The planning techniques explored include (a) multicontact trajectory planning and (b) whole-body pose planning, highlighting their contact and state planning strategies. In addition, the pros and cons of categorized approaches in (i) sampling-based, (ii) optimization-based, and (iii) learning-based methods are summarized.

trajectory planning in a unified fashion, *a.k.a*, contact-implicit planning (CIP) [92], [93]. The primary challenge of this CIP lies in its high computational burden and combinatorial complexity of identifying potential contact mode sequences.

State-of-the-art procedures select planning algorithms for the underactuated system with two main categories: (i) optimization and (ii) learning, as illustrated in Fig. 5.

A. Optimization-Based Contact Planning

Unlike search-based contact planning, which primarily checks kinematic feasibility for expansion and often requires additional lower level planning to generate dynamically feasible motion, optimization-based contact planning [92] offers the possibility of *simultaneous* planning of whole-body motion and contact interactions, as illustrated in Fig. 5. This approach integrates dynamics directly into the contact planning process, eliminating the need for a hierarchical structure. A contact-implicit trajectory optimization (CITO) is formed by incorporating contact dynamics into the trajectory optimization (TO) formulation, allowing the solver to determine the contact modes, contact forces, contact positions, and whole-body motions all at once [94], [95], [96].

The recent improvements in computation efficiency through accelerated solvers have even enabled the use of CITO in a model predictive control framework for real-time planning on quadruped robots [97], [98] and robotic arms [99], [100]. However, for the humanoid robots, applying CITO to loco-manipulation has yet to be achieved.

B. Learning-Based Contact Planning

Learning-based approaches have demonstrated promising potential in planning multicontact tasks, such as using RL to generate velocity commands and contact sequences for quadruped locomotion [101], [102]. These learning-based planners are mostly

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

modular, making it possible to form a hierarchical architecture with model-based planners and controllers at the low level.

In addition, learning-based approaches can assist contact prediction, which allows additional information for contact (re)planning in real time. Precise contacts are often hard to obtain from motion capture data, making it challenging to learn directly from data. Naive supervised learning often yields implausible motions, such as object movement without contact or significant body-object penetration. To address this, [103] introduces contact correction by predicting motion relative to contacts, while Li et al. [104] first predicted object contact points and uses them to constrain whole-body motion synthesis. These methods show promise as loco-manipulation planners for humanoids. Similarly, Bahl et al. [105] learned affordances from video scenes—identifying potential contact points that can guide subsequent motion planning.

C. Challenges in MCP

Despite the advancements in MCP, several key challenges remain. First, computational efficiency remains a fundamental bottleneck. Contact planning inherently involves a combinatorial explosion in possible contact mode sequences, which poses major scalability issues. Optimization-based methods, such as CITO, often suffer from local minima and sensitivity to initialization. Learning-based approaches, though computationally efficient, struggle with adaptivity and precision in dynamic contact-rich control. A promising direction lies in combining these paradigms to balance global reasoning, dynamic feasibility, and real-time performance. Second, most existing methods work around fixed contact mode sequences, which diverge from the unpredictable and rapidly changing nature of real-world contact interactions. Unexpected contact events, which are often hard to observe due to visual occlusion and lack of whole-body tactile sensing, pose serious challenges for accurate contact estimation, control, and online planning. To enable more adaptable loco-manipulation behaviors in uncertain environments, future research could explore planning, control, and learning strategies that are robust or even agnostic to small, transient changes in contact.

Key Takeaways: While significant progress has been made in humanoid MCP, future work should focus on developing more integrated approaches that combine the strengths of search-based, optimization-based, and learning-based methods. Specifically, addressing the computational complexity of CIP and improving real-time performance will be key. Future directions could explore hybrid approaches that incorporate efficient contact sequence generation/contact dynamics, apply contact-implicit constraints in real-time, and achieve learning-based contact prediction to enhance robustness and adaptability in complex loco-manipulation tasks. The readers are recommended to further read the survey on humanoid MCP [106].

V. Model Predictive Control for Loco-Manipulation

Optimization-based model predictive control (MPC) has advanced significantly in robotics. The advantages of its flexibility to define versatile motion objectives, rigorous mathematical formulations, and widely available solvers establish MPC as one of the most popular approaches to trajectory planning for locomotion and manipulation, as illustrated in Fig. 6.

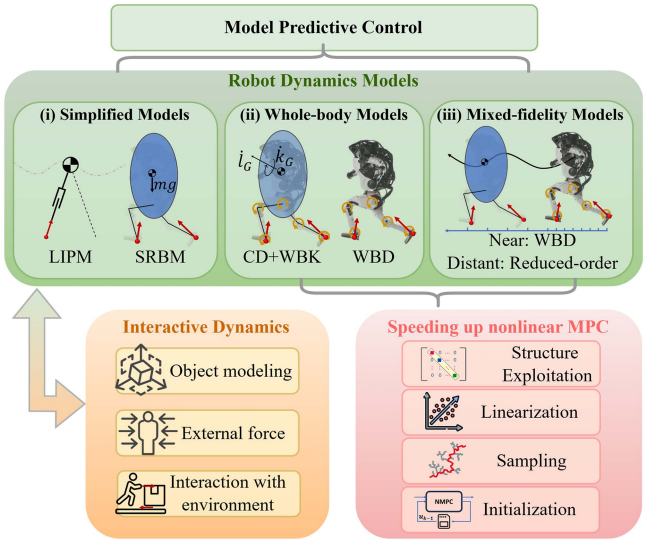


Fig. 6. Illustration of model predictive control in humanoid robotics, showcasing three primary categories of dynamics modeling in loco-manipulation: (i) simplified dynamics, (ii) nonlinear dynamics, and (iii) Mixed-fidelity dynamics. The figure also highlights the consideration of interactive dynamics modeling with environments and/or objects for loco-manipulation tasks. In addition, four common approaches to speeding up/simplifying NMPC are summarized.

A uniform optimization formulation of the loco-manipulation planning problem seeks an optimal state trajectory and control input over a finite future horizon. MPC is often formulated as an OCP.

Depending on the choice of dynamics models, costs, and constraints, the OCP formulation is commonly transformed as a linear convex MPC (e.g., [2], [107]) or a nonlinear MPC (NMPC) (e.g., [108], [109]). Table III summarizes recent MPC-based works on humanoid robots in loco-manipulation tasks.

A. Simplified Models

To enable high-frequency online planning with low computational cost, simplified, or reduced-order models (ROMs) are commonly used in MPC. The single rigid body model (SRBM), when linearized with predefined footstep sequences, enables convex MPC formulations [117] and has been used for dynamic behaviors like aerobatics [107]. In loco-manipulation, SRBM can approximate object interaction as external forces on the CoM, allowing for object-carrying over rough terrain [2]. Similarly, the linear inverted pendulum model (LIPM) is widely used for locomotion [26] and multicontact tasks [118]. While LIPM has been extended to loco-manipulation via teleoperation [112], its inherent limitations in modeling contact interactions require a lower level whole-body controller to handle balancing and manipulation.

B. Whole-Body Models

NMPC comes into play when constraints or cost functions become nonlinear, for example, dynamics constraints formed by kinodynamics and whole-body dynamics (WBD).

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RECENT MPC APPROACHES ON HUMANOID LOCO-MANIPULATION

Paper	Year	Robot Model*	Interaction modeling method	Locomotion (L) and Manipulation (M) Planning	MPC Frequency	Solving Method
[110]	2014	SRBM	Optimizing external wrench(es) at contact(s)	Unified	20 Hz	QP
[2]	2023	SRBM	Predefined external force	Unified	300 Hz	QP
[111]	2024	SRBM	Negligible object dynamics	Separated: L: MPC; M: Keyframe interpolation	–	QP
[112]	2019	LIPM	Negligible object dynamics	Separated: L: MPC; M: Teleoperation and retargeting	–	QP
[113]	2016	CD	Optimizing external wrench(es) at contact(s)	Unified	offline	SQP
[28]	2019	CD	Predefined external wrench	Unified	–	QP
[114]	2014	CD	Optimizing external wrench(es) at contact(s)	Unified	5 Hz/offline	SQP
[115]	2020	CD	Optimizing external wrench(es) at contact(s)	Unified	10 Hz	Interior-point
[109]	2023	CD	Estimated as external wrench through sensors	Unified	5 Hz	Interior-point
[116]	2021	WBD	Optimizing external wrench(es) at contact(s)	Unified	100 Hz	DDP

SRBM—Single rigid-body model; LIPM—Linear inverted pendulum model; CD—Centroidal dynamics; WBD—Whole-body dynamics.

In the context of humanoid motion planning, kinodynamic constraints are often referred to as the combination of CD and whole-body kinematics constraints [114], where CD is derived from the total momenta of the system, and captures the effect of full-body inertia of a multilinkage dynamics system [119]. For example, achieving consensus between CD and full-body kinematics in one TO generates versatile humanoid motions [114].

Leveraging joint-space WBD in MPC has gained popularity, particularly for free-floating articulated robots like humanoids. While WBD allows flexible contact modeling, its high nonlinearity and nonconvexity impose significant computational burdens on WBD-based nonlinear programs (NLP) or NMPC, making real-time planning challenging. This issue is especially critical for high-DOF humanoids in loco-manipulation tasks, such as payload transport, which require additional modeling of object dynamics and safety constraints. Thus, this section surveys methods to accelerate NMPC while preserving solution accuracy.

C. Mixed-Fidelity Models

Instead of using full joint-space dynamics across the entire horizon of an MPC, mixing multiple models of varying abstraction levels demonstrates improved performance and efficiency.

Cascaded-fidelity models (*a.k.a.* hierarchical dynamics) use different models to govern segments of the horizon [120], [121], [122]. These methods typically employ high-fidelity (e.g., full-order) models in near horizons and low-fidelity (e.g., simple) models for distant horizons, thus maintaining the solution accuracy in near horizons while solving the myopic issue by allowing a longer horizon using simple models. This approach could be suitable in loco-manipulation tasks as it would either simplify interaction dynamics as simple external forces or impose the object dynamics as part of CD in the far horizons to allow improved real-time computation compared to full dynamics models.

An alternative approach is to overlap multiple dynamics models across planning horizons, requiring consensus between models. For example, Budhiraja et al. [123] addressed this by decomposing the problem into two subproblems and alternating between them to achieve consensus. Similarly, Herzog et al. [124] alternated between CD and WBK subproblems. While model simplification remains effective for MPC [9], mixed-fidelity models offer higher fidelity but demand careful coordination across models.

TABLE IV
NMPC SPEED-UP STRATEGIES: ACCURACY, EFFICIENCY, AND LIMITATIONS

Method	Accuracy	Efficiency	Limitation and Challenge
i) *Dynamics modeling*			
Simplified model (Sec. V-A)	low	high	Possible infeasible motion trajectory due to simplification
Whole-body model (Sec. V-B)	very high	very low	Requires very accurate dynamics modeling of robot and object
Mixed-fidelity model (Sec. V-C)	medium	medium	Motion trajectory discrepancy due to model mismatch
ii) *NLP Structure Exploitation*			
Backward recursion:			
DDP [125]	medium	high	Poor constraint-handling
iLQR [126]	low	very high	
Derivative calculation:			
Auto-Diff [127]	very high	low	May lead to high memory usage
Hypergraph [128]	high	medium	
Exploring sparsity:			
FATROP [129]	very high	high	Requires specific problem structure
AdaptiveNLP [130]	high	high	Less effective in small-scale NLPs
iii) *Linearization*			
SQP [131]	high	medium	Scalability to high-order
Successive Linearization [132]	medium	high	Requires good initial guess
QP Speed-up:			
Condensed [133]	very high	medium	No state trajectory output
ReLU-QP [134]	high	very high	Only time-invariant states matrices
iv) *Warm Start*			
Gait Library [135]	high	very high	Constrained by onboard storage
Memory of Motion [116]	high	low	Requires large dataset coverage

Performance scale (low to high): very low→ low→ medium→ high → very high.

D. NMPC Speed-Up

In addition to tailoring dynamics modeling strategies for specific loco-manipulation setups, solving NMPC can be accelerated using several key methods. In the following, we summarize these methods and provide Table IV to compare their solution accuracy, real-time efficiency, and application limitations.

NMPC speed-up via structure exploitation: NMPC problems often involve complex dynamics and constraints that can be computationally intensive to solve. Exploiting the structure within these problems can significantly enhance their solvability and efficiency, such as extracting variables that directly interact with each other, identifying repetitive and symmetric structures, and arranging block-diagonal structures.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

NMPC speed-up via linearization: Another way to tackle the computational burdens of NMPC is through successive linearization, which involves linearization at every timestep around the nominal system state and control input. The linearized dynamics become piece-wise affine, which can be formulated in a large, sparse quadratic program (QP) and can be solved online [132], [136]. Aiming for high efficiency, rectified linear unit - quadratic programming (ReLU-QP([134], a GPU-accelerated QP solver, has improved convex MPC real-time control frequency in high dimensional space balancing tasks from the original 206 Hz to up to 2600 Hz.

NMPC speed-up via warm start: The real-time requirement motivates many researchers to seek a more effective initialization. One simple yet effective approach is to warm start with the solution from the previous iteration. Another common approach is to offload the computation burden from online to offline, e.g., the *gait library* [135]. It can be regarded as a specific type of warm-start technique and requires only a cheap online interpolation among the gaits to obtain an approximately optimal full-body trajectory. However, the key challenge lies in the management of massive trajectories with limited storage. In Section VII-E, we discuss a solution from the learning community: learning compact models to distill from large-scale offline trajectories.

E. Environment and Object Interaction Models for Loco-Manipulation

In this section, we survey motion planning algorithms for loco-manipulation tasks that involve interactions with the environment and/or objects with large weights and sizes, specifically in the context of loco-manipulation MPC. The existing loco-manipulation MPC algorithms differentiate the interaction with a fixed environment and a manipulated object.

1) Interaction With a Static Environment: An environment, including static surfaces such as the ground and walls, provides contact forces that contribute to the robot's stability and enable interactive tasks such as walking and pushing. Since the environment is static, the robot does not need to consider the environment's state or stability during planning. Instead, the robot is often required to deal with acyclic contact patterns and noncoplanar contacts given specific environment geometries. This challenging problem is referred to as MCP [106], [137]. In this section, we only discuss the motion planning aspect of the MCP in terms of multicontact MPC.

2) Modeling Interaction With a Manipulated Object: Given the diversity of tasks, creating a unified model for the robot-object system is essential. We introduce two common approaches to incorporate object dynamics into the MPC-based planning process [118].

The first approach models the manipulated object as external wrenches and plans the control input to compensate for them [2], [53], [115]. This approach offers a flexible solution, as it integrates well with the MPC regardless of the linearity of the model, which treats all contacts as external wrenches. However, the contact wrench needs to be predefined, for example, to compensate for the gravity of the object or to exert a user-specified pushing force. Static/quasi-static assumptions are usually made to neglect the dynamics of the object, resulting in less dynamic loco-manipulation motions.

The second planning approach incorporates the object's dynamics directly into the robot's dynamic equation of motion, creating a unified robot-object dynamic system [118], [138]. This approach eliminates the static/quasi-static assumption from the first approach and leverages the time-varying robot-object dynamics in MPC to achieve more dynamic and adaptive loco-manipulation behaviors. In such planning problems, the interaction wrenches are usually treated as control variables, and contact stability constraint on the interaction wrenches is enforced to securely attach the object to the robot. The planner generates the combined motions for both the robot and the object, leading to their desired states. Compared to modeling objects as external wrenches, this method requires a perfect knowledge of the object's state, which is more challenging from the sensing perspective.

Key takeaways for model predictive control: With the advanced capabilities of gradient-based numerical optimization in motion planning, MPC is gaining popularity in humanoid loco-manipulation, showcasing numerous variations in recent years of the literature. The essence of this method lies in making reasonable choices regarding the dynamics model, constraint, task definitions, and computation requirements. These choices often require expert design and tuning to tradeoffs among task versatility, solution feasibility, and optimality. By identifying the computation intensities and proper dynamics representation of the loco-manipulation tasks, one can offset the computation load by introducing simplified models and relaxed constraints in MPC. In addition, the MPC efficiency can greatly benefit from proper solver choices, an evolving area presenting opportunities for research on both solver-level and problem-formulation-level innovations. For further reading, we recommend the survey on MPC for legged and humanoid robots [13]. In addition, loco-manipulation tasks present further challenges due to the complexity of dynamic interactions with both the environment and objects, which leaves open questions on how to choose and formulate the interactive dynamics effectively based on the specific task requirements in an MPC setting.

VI. Whole-Body Control

Whole-body control (WBC) represents a body of controllers that generate joint torques, constraint forces, and generalized accelerations to achieve a given set of desired dynamic tasks [7]. Three common cases necessitate a computationally efficient WBC that can track desired trajectories and send torque commands to a physical robot.

1) The desired trajectory is computed based on an ROM. Such a trajectory encodes only an important subset of the robot's full-body motion (e.g., desired CoM and end-effector trajectories in operational space [139]) and does not contain information for all joints.
2) The trajectories are planned with a full-order model but are too computationally heavy [135] to be used in real

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

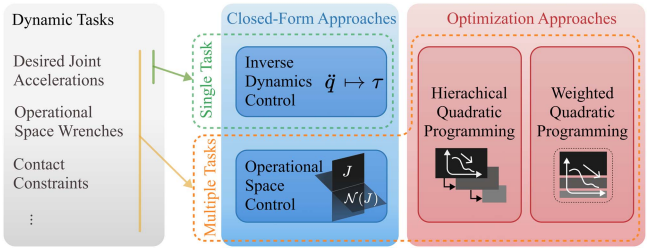


Fig. 7. Whole-body control is categorized into closed-form approaches and optimization-based approaches. Both approaches can incorporate multiple dynamic tasks and resolve task conflicts.

time, particularly for humanoids in loco-manipulation scenarios.

3) Environmental uncertainties and planning inaccuracies induce disturbances that require robust WBCs to compensate [140].

Therefore, the WBC has been widely used in the humanoid community.

A WBC is different from an MPC in two aspects. 1) The MPC often solves a receding horizon problem, whereas the WBC solves an instantaneous control problem (i.e., only for the current timestep). 2) The model adopted by MPC may vary based on the user choice, as discussed in Section V, but WBC usually employs full-order Euler–Lagrangian dynamics.

The main idea of WBC is that linear equations are sufficient to describe a universal set of dynamic tasks for locomotion and manipulation. The source of these dynamic tasks varies and may be predefined, computed online (e.g., from an MPC [139], [141]), or commanded through teleoperation [142], [143], [144], [145] and virtual reality (VR) interfaces [146], [147]. As shown in Fig. 7, WBC approaches can be categorized based on closed-form or optimization-based approaches to achieve a desired list of dynamic tasks.

A. WBC in Closed Form

An inverse dynamics controller is among the early works that address the WBC problem in closed form. In particular, it solves a single dynamic task: achieving the desired generalized acceleration $\ddot{\boldsymbol{q}} = \ddot{\boldsymbol{q}}^d$. The closed-form torque $\boldsymbol{\tau}$ can be analytically solved through dynamics projections [148], [149], [150].

In addition, a set of operational-space tasks and constraints can be achieved given the redundancy in the degrees of freedom of humanoid robots. As a multitask example, a humanoid robot is often tasked to generate interaction forces with a low priority while maintaining the whole-body balance with a high priority. Operational-space control (OSC), *a.k.a* task-space control, achieves multiple dynamic tasks by prioritizing tasks in a hierarchy [148], [151]. Tasks with low priorities are solved within the null space of the high-priority tasks, enforcing that hierarchies are strictly maintained among tasks. Such a task hierarchy is also named the stack of tasks [152].

Overall, closed-form approaches are computationally efficient and are straightforward to implement. However, they have difficulty in incorporating inequality tasks, such as joint limit and obstacle avoidance. Although this issue can be addressed within the closed-form approach, such as using a smooth operator [152], much of the community uses optimization-based methods that address this issue efficiently.

B. WBC Through Optimization

In contrast with closed-form approaches, there have been a variety of studies formulating the WBC as an optimization problem. These optimization-based methods enhance the flexibility of WBC, enabling the modular addition and removal of dynamic tasks [153], [154], including inequality tasks.

A salient feature of optimization-based WBC is the resolution of conflicting dynamic tasks through two prioritization schemes: strict task hierarchy [155] or soft task weighting [113], [156], [157], [158].

A strict task hierarchy can be ensured through a cascaded hierarchical QP. This method sequentially solves a series of QP subproblems with tasks from high to low priorities; lower priority QPs produce solutions within the combined null space of all preceding QPs [155]. However, solving multiple QP subproblems imposes significant computation and memory burdens. In addition, the hierarchical QP inherits a common issue of OSC: the task Jacobians becomes rank-deficient when approaching singularities, which induce large unstable movements [159].

In contrast to hierarchical QP, which solves tasks in a strict order, weighted QP resolves task conflicts by treating dynamic tasks as *soft constraints* within the cost functions, with weights indicating their relative priority. Therefore, weighted QP can be regarded as a special case of hierarchical QP with only one hierarchy or vice-versa, as documented in [160]. Such a setup enjoys the benefit of solving only a single optimization, which is faster than a hierarchical QP and can be further accelerated by exploiting sparsity and warm-start. However, tuning weight parameters can be burdensome for a large number of tasks, and can lead to instability [161]. Even with well-tuned parameters, the loss of strict task priorities means low-priority tasks can interfere with high-priority ones. Nevertheless, weighted QP is still widely applied in many robotics studies due to its easy setup and computation efficiency compared with hierarchical QP [113], [157].

C. WBC for Loco-Manipulation

WBC for loco-manipulation aims to achieve the desired motion while maintaining instantaneous *balance* and *contact stability*. Given the desired motion and contact sequence, loco-manipulation control can be categorized into two folds. 1) When all interactions with the environment and objects are static or quasi-static, they can be modeled as external wrenches. In this case, the WBC solves a robot balancing problem with the external wrenches as dynamic tasks. 2) When the manipulated object has a substantial mass or is dynamically moving, such as carrying a heavy box, the object becomes an integral part of the robot-object system. Therefore, the WBC must account for the balance of both the robot and the dynamic object.

1) Interaction as an External Wrench: In this first category, a subset of contacts is responsible for interacting with the environment or objects to apply a desired wrench. This desired wrench

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

can be specified by a user or derived from the estimated object weight. Considering the desired wrench from the interaction, the remaining contacts maintain the system balance using two distinct strategies.

The first strategy involves simultaneously optimizing contact wrench, joint acceleration, and joint torque using the robot's full-body dynamics, as detailed in Section VI-B. In this approach, the desired wrench for interaction is a dynamic task within the WBC. The WBC must also satisfy the dynamics constraint [119], contact stability constraint [162], and balance stability constraint [163].

The second strategy, known as preoptimization [3], [164], involves two stages in sequence. First, it determines the optimal distribution of contact wrenches based on the desired rate of centroidal momentum derived from the balance stability of CD. The second stage computes the joint torques needed to realize the contact wrenches using inverse dynamics of full-body dynamics. Note that, deriving the desired rate of centroidal momentum in the first stage is particularly challenging due to the nonholonomy [165] of angular momentum, i.e., the kinetic momentum of rotation is not directly related to the orientation of body links. As a result, the body orientation requires additional regulation (e.g., joint-level postural feedback [3]) beyond simple feedback on angular momentum.

2) Interaction as a Unified Robot-Object Model: A unified robot-object system can leverage the additional object to regulate the robot's dynamics. This yields more dynamically feasible behavior when carrying heavy or dynamically-moving objects. The unified model incorporates each manipulated object as an additional "robot"—either a passive object or a real robot—and connects the robot and object via action-reaction force pairs [166]. The balance stability must consider the combined CoM and inertia of the robot-object system [167]. In addition, the contact stability between the robot and the object is maintained to ensure that the object remains securely attached. While direct control of the interaction forces is feasible, adaptive force control that regulates the relative position between the object and the robot offers greater robustness. This approach mitigates the impact of inevitable inaccuracies in modeling inertia parameters and stiffness properties [118].

Key takeaways for whole-body control: The core of whole-body control lies in addressing an inverse dynamics problem to produce joint-level torque control. However, this problem is challenging due to the underactuation and contact-constrained nature of humanoid robots. Closed-form approaches such as inverse dynamics control and OSC are computationally efficient. Therefore, they have been traditionally prevalent. On the other hand, optimization-based strategies, particularly QPs, are increasingly favored as they adapt more effectively to a wide range of task specifications and offer more reliable solutions. Undoubtedly, both lines of WBC research have significantly advanced the progress of humanoid robot control over the past two decades. In the near term, optimization-based WBC will continue to be a popular choice for low-level control to achieve high-level loco-manipulation tasks. We also see neural WBC [168], [169], [170] gaining popularity, as we will discuss in the following section. For further reading, we recommend the survey on optimization-based WBC for legged robots [9] and the chapter on closed-form WBC techniques for humanoid robots [7].

D. Challenges in Numerical Optimization

Robotic planning and control techniques that are formalized as numerical optimization problems heavily rely on advances in applied discrete mathematics and optimization theory. These advancements address challenges such as nonconvexity, numerical robustness, and real-time resolution performance. However, a plateau may have been reached in the transfer of these techniques to the field of robotics at large—humanoids specifically—despite exploiting the unique properties of robotic models. These properties can enhance the efficiency of optimization problem formulation and its resolution by tailoring them to specific applications. However, as evidenced by the formulation OCP, including MPC (see Section V) and WBC (see Section VI), inherent physical uncertainties can disrupt closed-loop performance. Extending these formulations to loco-manipulation primarily involves (i) augmenting models to incorporate loco-manipulated counterparts and (ii) refining contact models formulations to account for various interactions (e.g., impact [171], rolling, deforming). However, these extensions risk overcomplicating the problem, potentially hindering effective formalization, even if the resulting formulations are sparse.

Importantly, the contact-explicit formulations have the significant limitation of requiring the user to determine the contact mode sequence for the problem, which generally limits the ability to generate complex motions. Alternatively, contact-implicit formulations introduce complementarity conditions to eliminate the strict dependence on the contact mode sequence [95], [172], [173]. However, contact complementarity conditions are nonsmooth, introducing severe computational challenges. Generally, this is tackled via regularizing the complementarity problem, e.g., [99], which approximates the constraint with a continuous affine function. Even with this linearized approximation, contact-implicit approaches struggle to scale to the high dimensionality of humanoid robots due to excessive computation and numerical difficulty. However, improved solver formulations, either as first-order solvers such as alternating direction method of multipliers [174] or specialized methods [97] for handling the complementarity conditions, have the potential to increase the tractability of real-time contact-implicit optimization.

Possibly more importantly, all these approaches, however, still have only *local* optimality guarantees. The level of nonconvexity or nonlinearity of the problem influences the solution quality, which often results in poor local minima or suboptimal solutions. When the structure of the problem requires deviations from local candidate solutions, a feasible solution, even one that is only locally optimal, may never be found. This feasibility issue remains an open challenge for nonconvex nonlinear optimization.

VII. Learning Loco-Manipulation Skills

Robot skill refers to the ability to use its own perception, planning, and control capabilities to complete specified tasks autonomously [175]. Among a variety of robot skills,

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

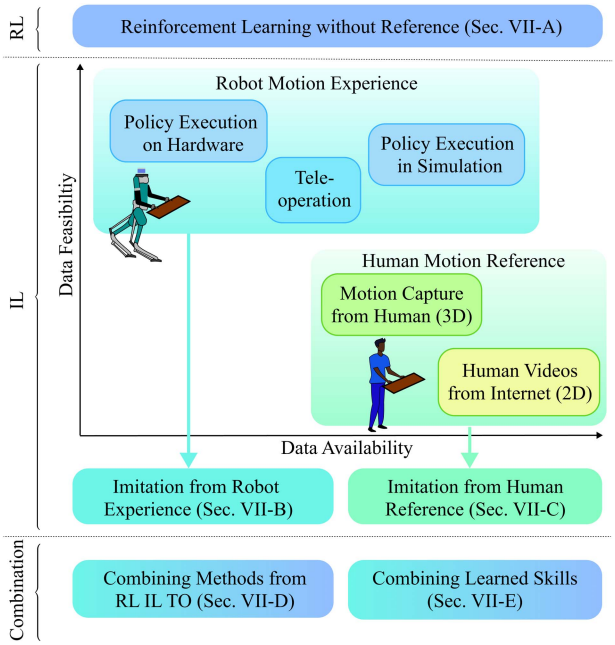


Fig. 8. Organization of approaches for skill learning. RL does not require reference data in a standard setup. IL makes use of four different data sources for efficient learning. The larger the morphology gap or sim-to-real gap is, the more challenging it becomes to learn effectively from these data.

loco-manipulation is highly valuable for augmenting and complementing human capabilities. This section reviews learning-based approaches that explore two main directions: 1) enhancing a specific skill in terms of agility, robustness, and safety, and 2) broadening the overall skill set of robots, revolving around two pivotal goals: versatility and generalization. Versatility refers to the capability of a single framework or policy to acquire *multiple* skills, whereas generalization involves adapting existing skills to new, out-of-distribution tasks and environments. Versatility can be viewed as a prerequisite for generalization. For a model to generalize across a wide range of tasks, it must possess a broad and versatile set of skills that can be applied in novel contexts. However, possessing versatile skills alone does not guarantee generalization—the model must also be capable of correctly prompting and adapting those skills to meet novel objectives. For instance, in the context of skill learning, a versatile policy can track multiple trajectories within the training dataset, while a generalizable policy should be able to track motions beyond those seen in the dataset, thereby demonstrating out-of-distribution capabilities.

Among learning-based methods, RL without demonstration and learning from demonstration, also known as IL, have shown remarkable proficiency for robotic skill learning. For the basics of RL and IL, we refer readers to the survey paper [15]. In this section, we discuss these methods for learning humanoid loco-manipulation skills. As shown in Fig. 8, we introduce RL in Section VII-A and IL in Sections VII-B and VII-C. Thereafter, we discuss the benefits of combining model-based and learning-based methods in Section VII-D. Finally, we discuss methods for learning versatile skills with a single policy in Section VII-E.

A. Skill Learning: RL From Scratch

RL, enhanced by modern deep learning toolchains and algorithms, has garnered significant attention in the field of robotics over the past decade. RL promises an effective way to learn motor skills by rewarding desirable behaviors and penalizing undesired behaviors, with minimal or no supervision during training. The end-to-end RL policies translate raw sensory input to actuation and are executable in real time.

RL provides distinct benefits but also comes with its own challenges. Compared with model-based methods, numerous RL methods are model-free [see Section VII-E for model-based RL (MBRL)], eliminating the need for accurate dynamics. Furthermore, RL does not require demonstration data, making its training setup straightforward. However, it often requires meticulous design of reward functions to shape the policy's behavior. In addition, deploying a learned policy on robot hardware often encounters a sim-to-real gap, a well-known issue induced by the inaccurate physical model used by the simulator. Policy learning from scratch often requires extensive and time-consuming interactions with the environment without a guarantee for task completion. For example, popular RL algorithms such as proximal policy optimization (PPO) [176] and soft actor–critic (SAC) [177] fail in most humanoid loco-manipulation tasks [178], partly due to the complexity of these tasks, the sample inefficiency, and the sparse reward design.

1) Comparison Between RL Algorithms: PPO [176], and similarly other on-policy RL algorithms such as asynchronous advantage actor critic (A3C) [179], is sample inefficient, but the policy update is fast and scalable. PPO is widely used in sim-to-real settings where simulation data are relatively cheap to obtain, e.g., in a parallelized simulation environment such as Isaac Gym [180]. SAC [177], and similarly, twin delayed deep deterministic [181] and deep deterministic policy gradient [182], are sample-efficient off-policy algorithms, but slower to train due to higher computational costs for the policy updates. In addition, Q-learning-based methods such as SAC can introduce model bias during the training process since they estimate gradients using a learned Q-function, whereas policy gradient methods such as PPO have the advantage of directly estimating the gradients of the learning objective. Off-policy algorithms are widely used when data collection is expensive, e.g., when the data can only be obtained from a real robot. The recent literature on RL for humanoid robots relies mainly on simulation data, and hence, PPO is widely employed. For other RL settings, such as real-world RL, we envision that more off-policy RL algorithms and novel RL algorithms will play an important role in learning humanoid skills.

2) Improving Learning Efficiency: Several approaches have been proposed to improve learning efficiency. Curriculum learning expedites training by allowing the policy to achieve simple tasks in the early stage of training and then progressively increasing the task difficulty and complexity [183]. Another approach is to promote exploration. Researchers use curiosity mechanisms, which encourage visiting unexplored states, to intrinsically motivate learning without an explicit reward design [184]. This has been shown to overcome the sparse reward setting and achieve

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

complex loco-manipulation behaviors such as door opening. Zhang et al. [185] also incorporated curiosity-based rewards to learn versatile loco-manipulation skills without any motion priors. Lastly, substituting reward terms with constraints in a constrained RL framework can significantly simplify reward tuning while achieving state-of-the-art locomotion performance [186].

3) Addressing the Sim-to-Real Gap: Sim-to-real is another formidable challenge in RL. Humanoid loco-manipulation faces steeper sim-to-real challenges. Humanoid robots possess higher DoFs and unstable dynamics, where the center of mass constantly moves out of the support polygon. Therefore, learning whole-body balance control is sensitive to parameters in a physics simulation, underlining the sim-to-real gap due to differences in dynamics between the virtual and real worlds. In addition, humanoids are expected to perform human-level manipulation tasks where the complex environment and the differences in observation space, such as visual appearances, aggravate the sim-to-real-gap. To address the sim-to-real challenge, a diverse set of mainstream approaches have been explored for humanoid robots. Domain randomization (DR) is among the most widely adopted approaches, which varies the properties of a robot model, such as mass, friction, and actuator dynamics, to train a generalized policy robust in the real world. Many humanoid works [37], [187] achieve sim-to-real transfer through DR. While DR is straightforward to set up, policy training is sensitive to the parameter randomization range, inducing laborious tuning: a larger range is challenging for the policy to fit all physical parameters (i.e., fail to learn), and a smaller range does not cover the full spectrum of parameters that can occur in the physical world (i.e., fail to transfer). System identification (SI) is another popular approach to enhancing model fidelity by approximating the system's input–output behavior from real-world data. Real-to-sim techniques use optimization [188] or search [189] to obtain simulation parameters that can best explain the real robot trajectories collected from policy execution. Learning to align simulation behavior with policy rollouts in the real world has also shown success in sim-to-real transfer [190]. However, it is challenging to collect real-world data covering the full space of states and actions, particularly for versatile, safety-critical tasks.

While SI uses real-world data to obtain an accurate model, domain adaptation (DA) uses real-world data directly to fine-tune a simulator-trained policy. In DA, the parameter distribution in simulation is defined as the source domain, and parameter distribution in the real world is defined as the target domain. The fine-tuning effectively transfers the policy from the source domain to the target domain. For example, sim-to-lab-to-real [191] develops a two-stage transfer: pretraining in simulation and fine-tuning in the real world. Despite these efforts to address the sim-to-real gap, a systematic solution remains elusive, as the aforementioned approaches are often case-specific. Against this backdrop, developing physics engines that facilitate real-to-sim construction and sim-to-real transfer is crucial. However, physics engines are constrained by real-time requirements and continue to face challenges in accurately solving contact dynamics. For instance, MuJoCo [192] employs a soft contact model that can lead to unphysical penetration artifacts, while PhysX [180] approximates friction cones using low-fidelity pyramids that widen the sim-to-real gap. Although existing solutions inevitably involve tradeoffs between accuracy and efficiency, advancements in contact modeling will significantly narrow the sim-to-real gap.

Key takeaways: RL provides an effective way to learn novel behaviors for humanoid loco-manipulation. However, in practice, the success of RL often relies on informative representations for both observation and action, extensive reward engineering, curriculum learning design, and a vast amount of trial-and-error experiences to estimate gradients. Consequently, using RL to train robots is almost never practical in the real world, at least for the current stage of development. Therefore, RL policies are predominantly trained in simulations. This makes the sim-to-real gap the Achilles' heel of RL, significantly dampening its initial promise. Compared with quadruped robots, the sim-to-real gap is particularly challenging for humanoid robots with high DoFs executing complex loco-manipulation tasks. This is why IL, leveraging limited but in-domain real-world data, has gained popularity over pure RL without demonstration data. For further reading in RL, we recommend the survey on learning-based legged locomotion [15] and the practical lessons for training robotic RL agent [193].

B. Skill Learning: Imitation From Robot Experience

IL is an umbrella term that represents a class of algorithms, including supervised, unsupervised, and RL that train policies from expert demonstrations. IL is particularly effective for complex tasks that are difficult to specify explicitly. Three essential steps exist in IL [17]. The first step is to capture the expert demonstration. The next step involves retargeting, where these demonstrations are mapped to the robot motions. If the captured motion comes from the same robot, such as from teleoperation, the retargeting step is unnecessary. The final step is policy training using the retargeted data.

We discuss four possible sources of demonstrations for humanoid robots:

1) policy execution;
2) teleoperation;
3) motion capture;
4) human videos, as illustrated in Fig. 8 and Table V.

We group these data sources into two categories: the robot experience data, which represents those directly obtained from the robots through policy execution or teleoperation, and the human data, which includes human motion captures and videos of human activities obtained from the Internet. Robot experiences exhibit smaller morphological discrepancies and are directly applicable to policy learning but are typically scarce. Conversely, human data are more abundant but present significant morphological differences to humanoid robots.

1) Obtaining Robot Experience Data: A reliable way to obtain robot experience data is to execute existing policies, either model-based or learning-based. However, collecting data on a physical robot requires a laborious setup of the environment and raises significant safety concerns. Therefore, conducting these executions in simulation is a more efficient approach, although

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE V
SKILL LEARNING METHODS BASED ON DATA SOURCE

Methods and Data	Pros and Cons	Algorithms
RL Without Reference	✓novel behavior ✗reward tuning	PPO [176], SAC [177]
IL Robot Execution	✓annotated data ✓dynamically feasible ✗scarce ✗limited diversity	Diffusion [194], IRL [195]
IL Teleoperation	✓multimodal behavior ✗rare full-body motion	BC-RNN [57], ACT [170, 196]
IL Motion Capture	✓accurate kinematics ✗small dataset ✗limited outdoor data ✗proprioception-only	RL motion imitation [197], GAIL [198], AMP [199, 200]
IL Human Video	✓diverse abundant data ✗non-physical motion ✗proprioception-only	RL motion imitation [201], GAIL [202], OKAMI [203]

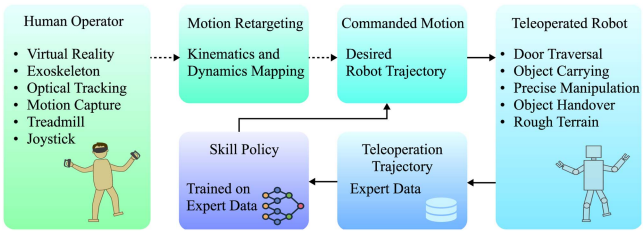


Fig. 9. Control flow for learning from teleoperated demonstrations. Expert data is created from teleoperated trajectories (dashed lines), which in turn is used to train and deploy an autonomous skill policy (solid lines).

the fidelity of the simulator will inevitably cause a sim-to-real gap.

Teleoperation is one of the most common ways to directly capture robot data commanded by human experts. A main advantage of teleoperation is its ability to provide smooth, natural, and precise trajectories for a wide range of tasks. Fig. 9 outlines the control flow of teleoperation data used as a source of policy training. The first step of this process is generating the demonstration through teleoperation, represented by the top path of the control flow (dashed lines). Motion retargeting maps human measurements from the teleoperation device to the desired trajectories in the robot domain. Robot execution data collected from teleoperation can be used to train autonomous policies (solid lines) that directly command the robot's motion without human intervention.

However, teleoperation for the *full-body* of a humanoid has a number of limitations. First, a majority of teleoperation systems capture only manipulation skills [204]. Generally, VR-based teleoperation schemes cannot sense the operator's gait and are restricted to simply commanding walking speeds and directions via joysticks [146], [205], [206]. Full-body sensing, including human gaits, often requires additional equipment such as IMU suits [144], [207] or exoskeletons [208]. These devices may be prohibitively expensive, bulky, complicated to maintain, and lack transparency and user-friendliness. Second, while teleoperation can generate versatile training data for a wide range of tasks, the utility of this data may be limited if the robot's kinematics fail to seamlessly retarget the reference human motion.

Although both teleoperation data and policy execution data are recorded from robots, they exhibit distinct characteristics. During teleoperation, human instructors tend to provide diverse demonstrations even for the same task. Therefore, teleoperation data are often multimodal, i.e., given a specific task, there exists a distribution of plausible ways to accomplish the task. In contrast, data from executing a single policy are often unimodal; that is, given an input, the output is often fixed. Different policy learning approaches have been proposed to address these multimodal and unimodal data features.

2) Approaches to Learning From Robot Experience Data: From unimodal policy execution data, which contains paired observations and actions, IL approaches are often used for policy distillation. Behavior cloning (BC) casts IL as a supervised learning problem, which remains one of the most straightforward approaches for robot skill learning [209], [210]. Another IL technique is inverse reinforcement learning (IRL), which reconstructs rewards from the data in addition to training an RL policy. The IRL study in [195] infers a generalizable reward of the expert demonstration for bipedal locomotion and then uses it to train an RL policy in unseen terrains.

To capture the data multimodality and produce diverse future actions, the action chunking transformer (ACT) [170], [196] is adopted to handle distribution shifts due to the compounding error inherent in naive BC. Diffusion policy [194], a BC method, shows the ability to acquire multimodal locomotion skills by learning from a large dataset collected from multiple expert policies. However, obtaining these skills requires large-scale versatile data, which motivates the scaling of data collection via teleoperation in industrial companies, e.g., Tesla and Toyota Research [204].

Key takeaways: Although collecting high-quality data demands considerable effort and resources, IL from robot experience remains a reliable method for attaining skills with expert-level performance. Industrial companies and research labs are increasingly focusing on scaling data collection to develop a broader range of diverse policies through IL. Especially, teleoperation is one of the most popular ways to collect humanoid robot experiences nowadays. For further reading on collecting robot experience data, we recommend the survey on humanoid robot teleoperation [211]. We also find the survey on IL of humanoid bipedal locomotion [17] a decent summary.

C. Skill Learning: Imitation From Human Data

While robot experiences can serve as a reliable data source, collecting loco-manipulation data directly from robots remains a formidable challenge. Collecting teleoperation data, even though it is one of the most commonly used approaches, is costly and tedious to scale. In addition, gathering robot data by deploying existing model-based methods or trained policies presents additional difficulties. Deploying these methods on hardware raises safety concerns. On the other hand, data collected from a simulator or synthesized from a model suffers from sim-to-real gaps. Furthermore, model-based methods based on human knowledge (e.g., dynamics models, heuristic trajectories) generate consistent but similar behaviors, leading to limited data diversity.

Learning from a large, diverse corpus of human data mitigates these challenges, as recording human data is more accessible

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

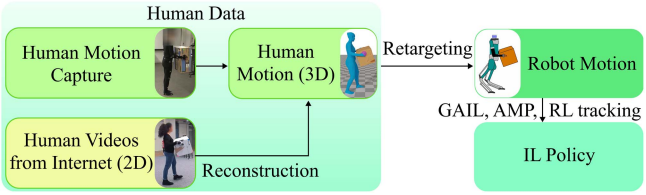


Fig. 10. Pipeline for learning from human data. The 3-D human motion data can be recorded from motion capture systems or reconstructed from 2-D human videos. Robot motion is retargeted from 3-D human motion for IL.

and scalable due to the reflexive usage of loco-manipulation skills by humans in their daily lives. Furthermore, training policies to imitate human data can simplify the synthesis of loco-manipulation behaviors. Recent research efforts in 3-D human motion data archival have surged in the vision and computer graphics communities. As shown in Fig. 10, there are currently two primary approaches to acquiring 3-D human motion data: (i) recording directly from motion capture systems and (ii) reconstructing from 2-D videos.

1) Obtaining Human Data: Various tracking systems are used to obtain human motion data. For instance, [104], [212] captures humans interacting with various objects while moving around. The following datasets, CMU [213], SFU [214], LAFAAN [215], and AMASS [216], are commonly used because they provide a wide variety of human motions. However, motion capture data often require heavily instrumented environments and actors, making them expensive to scale, and indoor lab settings provide little exposure to outdoor activities.

Alternatively, videos and images obtained from the Internet offer a rich and diverse source of human motion data, including athletic performances, artistic dances, or daily chores. However, motion extracted from internet data is usually of lower quality, containing noise, jittery, and nonphysical artifacts due to occlusion and motion blurs. Therefore, the reconstruction of accurate 3-D human poses from 2-D data remains an active research topic in the computer vision community.

Given the human dataset, researchers have been pursuing *motion generation*, leveraging human data to generate diverse and realistic motions. Recent advances in generative models, particularly diffusion-based approaches for video generation such as stable video diffusion [217], have significantly propelled progress in motion generation. A comprehensive survey [218] highlights the impressive capability of these models to generate realistic human motions. However, our focus here is training IL policies to generate *physically plausible* actions that achieve the motions demonstrated in human data. The training pipeline is shown in Fig. 10.

2) Challenges in Learning From Human Data: Using offline human motion data (from any source) to train humanoids inevitably creates an embodiment gap both in observation and action spaces due to different body proportions, joint configurations, and mass distributions between most humanoids and humans. Closing this embodiment gap requires retargeting, which involves mapping the motion collected from a source skeletal model to a target robot model. Previous work in computer graphics and robotics has explored various retargeting strategies, such as joint-space [144], [170], [220] and task-space correspondence [196], contact points [167], gait synchronization [207], [208], and motion feasibility filter [145], [169]. Developing systematic solutions for retargeting the entire human body, including dexterous hands, remains a critical topic for advancing humanoids.

Another correspondence problem is that the human data are only proprioception-based, which lacks sensory input and action output. Notably, these human data lack tactile or force measurement from interactions, which limits the capability of learning for loco-manipulation with rich physical interaction. To solve this issue, IL trains control policies that track references within a physical simulator. The physical simulator plays a key role in providing sensory input and validating the physical feasibility of the policy action.

3) Approaches to Learning From Human Data: Examples of learned human-like motions include walking using generative adversarial imitation learning (GAIL) [198], [202] and adversarial motion prior [199], [200], [221], which is an extension of the GAIL. Recently, RL-based motion imitation using motion capture data has successfully transferred to humanoid robots [169], [170], [219], [221]. Human motion imitation can also achieve robust policies capable of rich interactions with objects in unstructured environments. For example, the authors in [201] and [222] demonstrated the learning of full-body gymnastic skills for humanoids in physics simulation, utilizing video reconstruction data and RL-based motion imitation. The authors in [223] and [224] learned loco-manipulation policies enabling simulated humanoid characters to carry boxes. Furthermore, Wang et al. [197] mimic motion capture data to achieve tasks such as playing basketball and grasping objects. However, many of these approaches rely on privileged information, such as the ground truth of object and ego poses, from the simulator, which limits their extension to real-world hardware.

Key takeaways: Transferring skills from humans, especially through Internet-scale datasets, unlocks a broad range of loco-manipulation capabilities for humanoid robots. Although learning from human data holds great promise, a large body of what has been achieved in simulation has yet to be realized in real-world robots. Developing affordable and capable humanoid robots and high-fidelity simulators of real-world loco-manipulation scenarios can accelerate progress in this research direction. As we witness an increasing accessibility of humanoid robots in the market, we foresee that imitation from human data will enable humanoid robots to acquire a vast and diverse skill set. This skill set is fundamental to building humanoid FMs discussed in Section VIII.

D. Skill Learning: Hybrid Methods

Approaches that combine learning-based methods (IL and RL) and model-based methods are illustrated in Fig. 11.

1) Combining Pure RL and IL: The combination of IL and pure RL without demonstration data has led to effective sim-to-real transfers. A two-stage teacher–student paradigm is widely adopted [35], [37], [225]. In these works, a teacher policy is

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

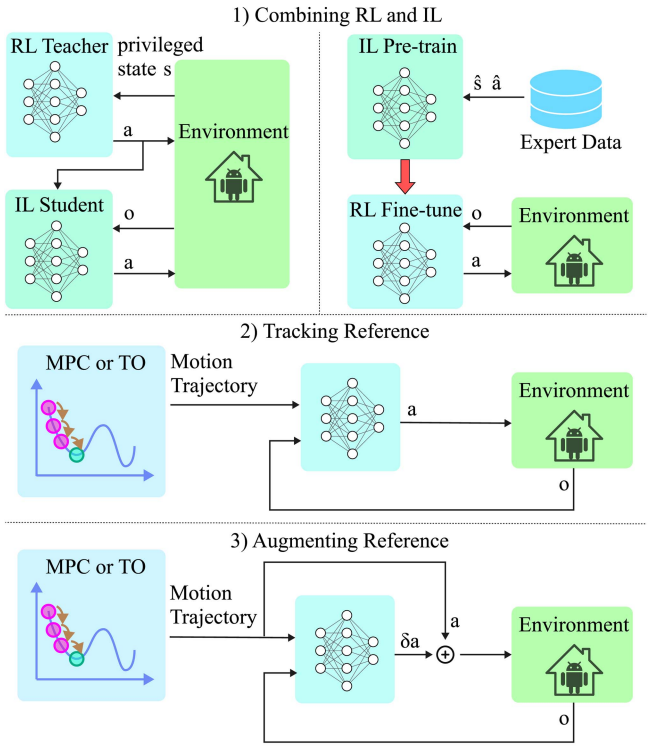


Fig. 11. Frameworks of combining model-based methods and model-free methods for skill learning.

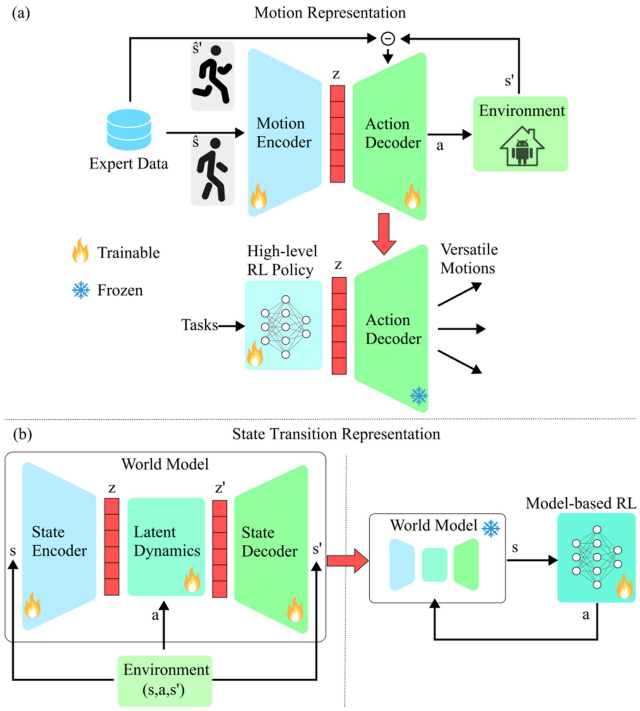


Fig. 12. Implicit skill representations. (a) Represent motions in latent space. (b) Represent state transitions with a world model and then use the learned world model for sample-efficient training.

first trained from simulated privileged observation using pure RL. Then, a student policy clones the behavior of the teacher, achieving a similar performance using only partial observations. The trained student policy is readily deployable on hardware with onboard observations. Another two-stage paradigm [226] reverses the order of the two policies. It first uses IL to pretrain an imitation policy from expert data and then uses an RL policy to fine-tune the imitation policy, achieving performance beyond the IL expert and adapting to different environments or tasks [227].

2) Learning to Track Trajectory Reference: Combining model-based methods with learning-based methods is extensively explored. Numerous works on quadruped robots [228], [229] use MPC to generate reference motions and use them as imitation rewards for RL-based motion imitation. However, calculating MPC online can prolong the training time and occasionally fail to find a feasible solution. Online RL that tracks *offline-generated* trajectories avoids this problem and has achieved effective RL policies for versatile bipedal locomotion [33], [230], [231]. Supervised learning is also used to mimic TO-generated offline motions. For example, Kwon et al. [232] allowed humanoid robots to achieve brachiation on monkey bars. In addition to enhancing learning methods with model-informed trajectories, a learned policy can also suggest effective warm starts or hyperparameters for model-based methods, thus significantly reducing the iterations required for convergence to an optimal trajectory.

3) Learning to Augment Trajectory Reference: Rather than tracking the reference trajectory, augmenting the reference with a residual is another popular approach. Early work on dynamic movement primitives [233] modifies the reference trajectory by learning a task-specific force output, which achieves a humanoid racket swing. A milestone in trajectory augmentation is the demonstration of bipedal locomotion in [234]. Since then, there have been more studies on trajectory augmentation with successful dynamic locomotion [235], [236]. Recently, loco-manipulation skills are also achieved through augmenting TO-based reference trajectory [237].

Key takeaways: This section summarizes hybrid methods with successful humanoid hardware deployments. Hybrid methods that combine learning methods with and without demonstration are effective for learning complex and robust behaviors. We also show that model-based methods are not in conflict with learning-based methods, as they can provide guidance for efficient learning. Overall, hybrid methods that combine learning-based and model-based approaches have achieved superior efficiency, versatility, and performance in humanoid tasks that outperform single methods alone.

E. Representation and Composition for Versatile Skills

A good representation of skills makes it easy to compose tasks. In this section, we explore structured representations for skill composition to achieve versatile and multiskill tasks. Structured representations improve memory efficiency and allow a *single* policy to achieve multiple tasks. Next, we introduce three well-received structured representations: motion representation, goal representation, and state transition representation, all shown in Fig. 12.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

1) Motion Representation: Motion representation extracts the essential features and temporal dependencies of high-dimensional long-horizon motions [238]. Specifically, motion representation encodes high-dimensional motions in a low-dimensional latent space. Such latent-space representations are commonly learned in an unsupervised manner using generative models such as variational autoencoders [239] and generative adversarial networks [240]. The result of learning motion representation is often a model that can synthesize versatile motions given latent codes. Therefore, the generative model can be reused for new downstream tasks by pairing it with a high-level task-specific policy. For example, several studies [241], [242], [243] learn a high-level RL policy to efficiently use a reduced-dimensional latent space motion representation and allow simulated humanoid robots to follow a set of user-commanded tasks.

2) State Transition Representation: The latent space can also represent the transition dynamics of a Markov decision process (MDP). In this representation, data collected from the MDP are used to train a dynamics model that predicts the transition probability between abstract, compressed representations of MDP states. This learned dynamics model is often referred to as a world model [244]. Sampling from a world model yields imaginary data, which can be achieved efficiently in parallel. Leveraging the imaginary data, MBRL achieves greater efficiency compared to a typical model-free setting, where interaction data must be obtained from a simulated or real environment. MBRL has shown success in agile motor skills on humanoids [245], [246]. Another approach, TD-MPC2 [247], uses the world model in an MPC fashion, planning actions that lead to imaginary trajectories with high scores. Beyond data efficiency, the world model can mitigate the sim-to-real gap by fine-tuning a small batch of real-world data [248].

Key takeaways: Enabling robots to accomplish versatile skills and multiple tasks is one of the main trends in robot skill learning. Whereas obtaining and blending single-skill policies is widely explored, more recent methods put efforts into achieving multiple tasks in a *single* policy. This requires a more structured representation of the skill motion, the task goal, and/or the environment dynamics. Latent space models, goal-conditioned policies, and world models are promising approaches toward this objective. However, many of these methods are still limited to the computer graphics community and yet to be implemented on humanoid hardware.

F. Learning for Humanoid Loco-Manipulation

Only a few studies have demonstrated sim-to-real transfer for loco-manipulation skills. A large part of these studies relies on sim-to-real RL. However, sim-to-real RL for loco-manipulation typically involves complicated reward designs that are fine-tuned for specific environments and tasks [249], [250] (see Section VI-I-A). To enable loco-manipulation tasks, these methods define the contact sequence either implicitly using reference trajectory [237], [249] or explicitly via reward design [185], thus increasing the success rate of sim-to-real transfer. To address the uncertainty of physical properties of the manipulated object, most RL approaches rely on DR to provide robustness to object parameters [37].

A hierarchical RL structure that manages distinct skills is a viable strategy for achieving various loco-manipulation tasks such as falling recovery and ball kicking [187]. However, hierarchical RL that relies on distinct skills has scalability issues. On the other hand, IL methods, particularly those employing RL with motion imitation, have enabled humanoid loco-manipulation via teleoperation, as outlined in Section VII-B. Although teleoperation is not autonomous, it is a crucial intermediate step to collect humanoid data. This has led to promising developments in *autonomous* loco-manipulation skills [169], [170], with the potential for further expansion to more diverse loco-manipulation tasks.

Key takeaways: Although learning-based methods for humanoid loco-manipulation are less developed than model-based methods, their significance should not be overlooked. Learning-based methods are potentially more robust, as they can adapt to unstructured scenarios that model-based methods struggle to explicitly address, such as recovering from a fall in an arbitrary configuration [251]. Moreover, learning-based methods can find emergent behaviors that are challenging for model-based methods [184]. The recent success of quadrupedal loco-manipulation [252], [253], [254] shows significant promise for humanoids. However, transferring these algorithms from quadruped to humanoid is challenging because of the more complex dynamics of humanoids, which require enhanced safety measures and precise balance control.

G. Challenge in Lack of Benchmarks for Loco-Manipulation

The development of humanoid loco-manipulation skills is still in its infancy compared to more established tasks such as humanoid locomotion [255] and tabletop manipulation [256]. To advance this emerging field, there is a pressing need for large-scale, systematic benchmarks specifically designed for humanoid loco-manipulation, encompassing both simulation and real-world environments. Benchmarks that define loco-manipulation tasks and evaluation metrics can significantly accelerate research in this field.

For benchmarks in simulation, HumanoidBench [178] and Mimicking-Bench [257] offer a list of loco-manipulation tasks, such as sitting on a piece of furniture and wiping a window. SMPLOlympics [258] presents a collection of sports as tasks for simulated humanoids. These benchmarks provide metrics for evaluating the performance of humanoid algorithms and help verify the reproducibility within MuJoCo [192] and Isaac Gym [180]. However, these tasks represent only a subset of human skills, and the evaluation metrics remain highly task-specific. Expanding the benchmarks to include a broader range of real-world tasks, along with more standardized metrics, would enhance their utility and practicality. Moreover, the sim-to-real gap continues to limit the practical value and reliability of these simulation benchmarks: it remains uncertain how well performance in simulation translates to real-world deployment, although promising results have been demonstrated in zero-shot sim-to-real transfer.

For hardware benchmarks, the development of affordable and capable humanoid robots as standardized platforms for

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

hardware evaluation can significantly accelerate the research. Research efforts on open-source humanoid hardware and software, such as Hector [2], the MIT Humanoid [107], the Berkeley Humanoid [259], and the Duke Humanoid [41] represent valuable contributions. More importantly, rapid prototyping of diverse humanoid platforms has become a powerful catalyst for advancing cross-embodiment generalization, while the mass production of standardized systems plays a pivotal role in enabling reproducible research and scalable benchmarking. A particularly noteworthy initiative is the robotic grand factory (RGF) project [1], which exemplifies this vision through the iterative development of the Q-series humanoids. The RGF stands out for its ambitious scope and practical impact—delivering multiple robot versions tailored for a broad spectrum of real-world tasks, including mobile phone charging, object organization and storage, and front-desk reception. This line of work significantly lowers the barrier to entry for large-scale deployment and benchmarking, marking a major step forward in the humanoid robotics field. Complementing these efforts, the construction of remote-accessible real-world environments, as exemplified by the DARPA Robotics Challenge and Robotarium [260], enables researchers to focus on advancing algorithms.

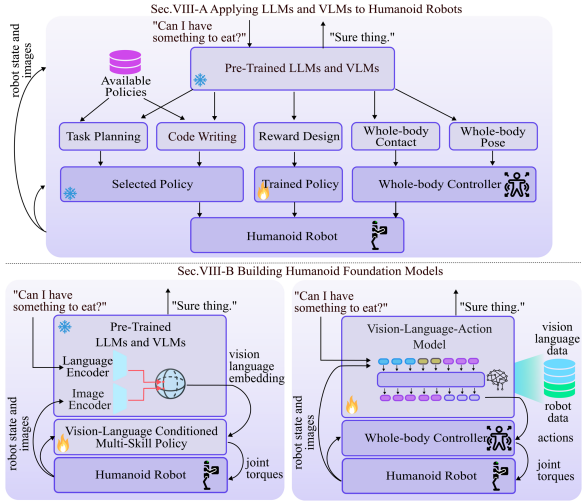


Fig. 13. Section VIII-A details different approaches to applying LLMs and VLMs in robotics tasks. These approaches prompt pretrained LLMs and VLMs to generate task-relevant intermediate representations that can be executed by low-level policies or controllers. Section VIII-B presents two approaches to building humanoid FMs. The first approach uses robot data to train multiskill policy, and the second approach fine-tunes an existing VLM. Both approaches directly output actions for low-level humanoid control.

H. Challenge in Data Scarcity

As discussed in Section VII-B, the four data sources present a tradeoff between quality and availability. The lack of high-quality large-scale robot data becomes a bottleneck for robot skill learning. To solve the bottleneck, much effort has been put into data scaling. There is ongoing debate about whether scaling is the best approach to generalist humanoid robots.

The central question we must answer is which aspect of human motion we want robots to learn. Some humanoid tasks can be achieved simply by mimicking 3-D human pose trajectories [169], [170], [203], [207], [219], but a true general-purpose robot emerges from purposive learning: the ability to identify meaningful intentions from human data and adapt past experiences to new tasks or environments [261]. Therefore, human data must teach the robot not only *what* humans are doing, but also *how* and *why* they are doing it. Current data acquisition methods that capture human joint poses only enable learning what humans are doing. In this regard, imitation of human data at the trajectory level is not fundamentally generalizable due to the inevitable gap in morphology and in the surrounding environment.

We argue that generalization in loco-manipulation is achieved by including the motion of the manipulated objects, enhanced with a greater variety of sensing modalities instead of data quantity scaling. To develop truly versatile and adaptive humanoids, human data should also include that of the manipulated objects, cognitive actions (e.g., trust, compete, collaborate) paired with multimodal observations (e.g., whole-body haptic sensing, egocentric images [262]), so that humanoids can learn the "how" and "why." However, instrumenting the environment and the manipulated objects with force and tactile sensing is highly complex and difficult to scale. Potential approaches include leveraging vision-language models to extract information from videos [263], [264], [265], employing video generation models [266] or physics-based simulation [180], [192] to generate synthetic data, and adopting hybrid strategies that effectively combine these synthetic data sources with limited real-world multimodal data [267], [268]. Recent work aims to bridge the gap between human animation and humanoid applications by collecting human multimodal observations [269], [270] and human kinetics [271]. Together with the rapid advancement in humanoid hardware, purposive learning with more informative human data will become the mainstream approach to achieve versatile and general-purpose humanoids.

VIII. FMs for Humanoid Robots

FMs are large pretrained models using Internet-scale data [272]. Recent progress in FMs such as large language models (LLMs) and vision-language models (VLMs) has demonstrated groundbreaking capabilities in solving a wide range of downstream tasks (through in-context learning or fine-tuning), such as code generation and video understanding [273]. The common sense reasoning capabilities of FMs have inspired many explorations of their applications in robotics [10], [11]. Strategies to leverage FMs for robotics can generally be categorized into two, as shown in Fig. 13. The first strategy (Section VIII-A) elicits actionable knowledge from pretrained LLMs/VLMs for robotic tasks, without additional model fine-tuning. The second strategy (Section VIII-B) collects abundant robotic data to fine-tune or cotrain a *robot foundation model* (RFM) that generalizes to control tasks with common sense reasoning capability [274], [275], [276], [277], [278], [279].

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

A. Applying LLMs/VLMs to Humanoid Robots

Applying LLMs/VLMs to humanoids is still a nascent field. Many studies have demonstrated the successful deployment of LLMs/VLMs across various robot embodiments, such as dexterous hands [280], manipulators [281], mobile manipulators [282], quadrupedal robots [283], and bipedal robots [284].

Among these works, a majority way of using LLMs/VLMs is to leverage pretrained models without robot data. Although these pretrained models have semantic understanding capability and context awareness, they often lack embodied knowledge and can prescribe actions that are ambiguous or nonadmissible. Therefore, considerable research efforts have focused on task-planning mechanisms to enable the generation of admissible action plans. For example, SayCan [282] ranks available actions based on value functions obtained during the training of corresponding action policies for mobile manipulators. VLM-PC [285] restricts GPT-4o to output plans with skills available only for quadruped navigation.

1) FMs for Selecting Low-Level Skills (Task Planning): Such task planning capability of FMs has advanced the complexity of tasks that a humanoid can accomplish. For instance, the authors in [286] and [287] used pretrained LLMs to select skills and task goals for animated humanoid characters. OmniH2O [169] employs GPT-4 [273] to select autonomous skills such as greeting a human. HYPERmotion [288] applies an LLM to construct task graphs that enable a hybrid wheeled-leg robot to execute complex loco-manipulation tasks. However, using FMs with a fixed skill set limits skill versatility. In addition, for complex behaviors such as those in humanoid loco-manipulation tasks, it is essential to allow FMs to author the detailed motions of low-level skills, instead of selecting from the existing, relatively-abstract low-level skills.

2) FMs for Authoring Low-Level Skills: Thus, many research efforts have focused on identifying the best bridge between FMs and low-level robot skills. For example, researchers have proposed to generate code [281], [289] and reward functions [283], [284], [290] as intermediate representations for bipedal and quadrupedal robots. Compared to selecting the existing skills, these intermediate representations provide additional flexibility in adjusting the generated motion. Furthermore, FMs can generate whole-body poses [291], [292] and whole-body contacts [293], [294] for humanoid robots. These techniques allow users to intuitively direct a robot's behavior through expressive inputs such as natural language, images, or even gestures.

B. Building Humanoid FMs

While most FMs are developed in the vision or language domain, building them in the robotics domain for embodied intelligence is a natural extension. Similar to LLM and VLM, RFMs are large models trained from Internet-scale robotic datasets. These RFMs often process multimodal inputs (e.g., egocentric images, and natural language as task description) and enable interaction with the physical world through robot action outputs. Therefore, they are also called vision-language-action (VLA) models. Leveraging the Internet-scale multimodal dataset, RFMs hold the promise of generalization across diverse tasks and provide a natural interface for human–robot interaction, both essential for real-world robot applications.

1) Hierarchical Vision-Language-Conditioned Policy: One popular approach to building RFMs is to create a multitask sensorimotor control policy using a high-capacity model that can consume large amounts of robot data. The trained policy can perform a wide variety of low-level skills, even for multiple robot embodiments. To enable high-level reasoning/planning, this approach often employs a hierarchical framework combining a pretrained LLM or VLM with the low-level control policy, initialized from scratch. During training, this setup aligns the semantic knowledge of the LLM and VLM with the physical behavior of the control policy, enabling cross-modal capabilities such as language-to-action. Transformer [295] is a common choice for such a low-level policy due to its scalability. To enable interactive user commands, the low-level policy is conditioned on language and/or image inputs, often tokenized with pretrained text or image encoders from LLMs or VLMs. For example, RT-1 [274] and VIMA [276] both train a language-conditioned visuomotor policy with a large amount of manipulation data and have demonstrated the ability to perform a wide range of skills. However, successful implementations are limited to robots that have stable dynamics and a large amount of high-quality data collected with significant resources [275], [277], [296], [297].

Applying such a hierarchical framework to humanoid robots is an exciting new direction. Recent works have extended RFMs to the humanoid upper body [267], [278], [279]. For example, NVIDIA developed GR00 T N1 [267], a general-purpose FM for humanoid robots, and Figure AI demonstrates Helix, a hierarchical VLA for dexterous manipulation and collaboration between two humanoid robots. However, building an RFM for humanoid loco-manipulation remains a challenging endeavor: the inherent instability in locomotion and high-dimensional action space in dexterous hands makes it difficult to collect high-quality data efficiently. Therefore, VLA for humanoid loco-manipulation is shown only in simulation. For example, HumanVLA [209] and SuperPADL [210] train a humanoid action policy aligned with the latent space of pretrained VLMs, enabling skills based on image and language inputs.

2) End-to-End VLA Model: To leverage prior knowledge in FMs, another popular approach is to build an end-to-end VLA model, as exemplified by RT-2 [275], OpenVLA [277], and Gato [298]. This approach treats robot data (i.e., observations and actions) as tokens in the language model's vocabulary, allowing direct fine-tuning or cotraining with existing VLMs. Unlike the previous approach, VLA outputs actions as tokens directly without relying on trainable low-level policies. The VLA model not only generates robot actions for diverse skills, but also retains semantic reasoning abilities in the language and vision domain, enhancing its generalization capacity compared to models trained in single domains. For example, RT-2 [275] represents the actions as a string of numbers similar to the tokens from the pretrained vision and language tokenizer of the base VLMs. However, representing actions with stringified numbers can be token-inefficient with high-DoF robots such as humanoids.

A key research question in building RFMs is the design of effective algorithms and model architectures. Although most of

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

today's RFMs are based on autoregressive transformer models [295], their computational inefficiency over long sequences poses a significant challenge for both training and inference. This has driven the exploration of alternative models that are both efficient and high-capacity, such as state-space models [299].

The key to the success of training an RFM, especially in the case of VLA, heavily depends on the choice of input and output representations. Most RFMs take as input a combination of task descriptions in language, visual observations of the surrounding environment, and the history of robot states. Outputs typically consist of robot actions, which are derived from either a multi-task policy (the first approach) or an end-to-end VLA model (the second approach). There are variations in the VLA model where its token outputs specify more than just actions. For example, Octo [300] and RDT-1B [301] use the token output for a diffusion denoising process. π_0 [296] maps a learned token to robot actions through a diffusion head, enabling high-frequency control (up to 50 Hz for a bimanual manipulator with a wheelbase). GR-2 [302] predicts tokens that represent future images and actions; thus, it functions as both a world model and a visuomotor policy. For humanoid robots, input and output are not yet well defined. For tasks involving rich physical interactions, the force feedback is as crucial as egocentric visual input. Given the unstable dynamics of humanoid robots, a more practical approach is to use end-effector and body poses as action outputs from the RFM and adopt additional low-level policies to ensure balance and safety via high-frequency feedback control. Incorporating robotic data as a new modality into state-of-the-art FMs would require significantly more data. Hence, designing effective input/output representation for humanoid loco-manipulation tasks still remains an open research question.

C. Opportunities and Challenges in FMs

Integrating FMs into humanoid robots offers distinct opportunities and challenges. On the opportunity side, since the majority of data used to train FMs is collected by humans, the knowledge embedded in these models is inherently biased toward human-like embodiments. Consequently, humanoid robots could potentially utilize existing knowledge in FMs more effectively due to a smaller embodiment gap. This advantage extends beyond planning and control capabilities to include interactions with humans using natural modalities such as language and gestures.

A major challenge in applying FMs to humanoids arises from the high inference cost. Running large FMs using only onboard computing is not feasible due to the limited power and computation, which hampers real-time hardware control. To address this challenge, several solutions have been proposed. One effective strategy involves adopting a decentralized hierarchy [278], where FMs operate over the cloud and provide only high-level decisions at a lower frequency, while another controller remains onboard and manages real-time task execution. However, the inference delay and internet latency may impede the control performance. The training of FM is also resource and time consuming. For example, training the LLaMA model took 34 days on 992 NVIDIA A100-80B GPUs [303], which incurs high cost, high energy consumption, as well as carbon dioxide emission. As FMs continue to scale up, the training cost increases further. A promising approach to maintaining a reasonable cost for training RFMs is to leverage parameter-efficient fine-tuning techniques. For example, OpenVLA leverages the LoRA technique to fine-tune an FM with a robotics dataset [277].

Beyond computational limitations, FMs face additional challenges when deployed in humanoid systems. Safety constraints are particularly concerning, as models trained on internet data may generate unsafe actions when controlling physical robots that are inherently unstable. Establishing a thorough evaluation procedure to assess the safety level of humanoid robots will be essential. Moreover, ethical considerations arise from biases present in human-generated training data, which can lead to unfair or discriminatory robot behaviors across different user demographics. These biases may be amplified in embodied systems that interact directly with humans in diverse settings. Federated learning approaches could help address some of these ethical concerns by enabling model training across distributed devices while preserving data privacy and potentially reducing centralized data biases [304].

Key takeaways: With rapid progress in FMs and humanoid robots, a plethora of research studies on humanoid embodied intelligence is anticipated. The first method outlined in Section VIII-A provides a more accessible way to leverage VLMs and, thus, has been the popular approach taken by researchers so far, but VLMs as is lack a deep understanding of robot actions. In the near future, we expect increased research efforts devoted to VLA models described in Section VIII-B to address this issue. It takes great effort to train RFMs from 1) a large amount of real robot data (e.g., Gemini-Robotics [278], [279]) or 2) a combination of robot data with synthetic data from simulation or video generation models (e.g., GR00 T N1 [267]). With the development of RFMs, we expect to have FMs that can well understand robot action and the physical world, enabling rapid deployment of robotic controllers achieved with little to no additional robot data. For further reading of FMs for robotics, we recommend the survey in [10] and [11]. Since most of the FMs are built with the Transformer backbones, please refer to [295] for a comprehensive mathematical foundation.

IX. Additional Discussions

A. Model-Based Methods Versus Learning-Based Methods

Based on the studies in this survey, we provide a clear comparison between model-based and learning-based methods in Table VI. In this table, we compare the performance in several aspects: the ability to incorporate multimodal sensor input (e.g., vision and tactile sensing), the control robustness, the motion accuracy, the real-time feasibility, the versatility, and the generalizability of each method. In addition, we evaluate the state-of-the-art capability in several humanoid skills, including locomotion, loco-manipulation, and whole-body multicontact control. We list representative papers that achieved these skills as supporting evidence for this evaluation. Note that many algorithm performances reported are anecdotal and highly depend on their implementation. Therefore, obtaining a concrete numerical comparison is challenging. This is mentioned in Section VII-G challenges in lack of appropriate benchmark.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON BETWEEN MODEL-BASED AND LEARNING-BASED METHODS

	Model-based	Learning-based		
	MPC, WBC	RL	IL	VLA
Performance measure				
Flexibility to incorporate multi-modal sensor	low	medium	high	very high
Robustness	high	very high	low	very low
Motion accuracy	very high	medium	medium	very low
Real-time feasibility	high	high	high	low
Versatility	high	medium	high	high
Generalizability	medium	low	low	high
Achieved performance of humanoid skills				
Locomotion	very high [6, 305]	very high [32, 35]	medium [194]	NA
Loco-manipulation (hand and foot contact)	high [2, 306]	medium [36, 185]	low [170]	NA
Whole-body multi-contact	medium [307]	high [308]	NA	NA

A key takeaway from this comparison is that learning-based methods exhibit superior robustness via RL and versatility via IL, while model-based approaches offer advantages in motion accuracy. The emerging paradigm of VLA holds promise for enhanced generalizability; however, its effectiveness in learning humanoid skills has yet to be convincingly demonstrated.

B. Hardware Limitations for Humanoid Robots

Advancements in control, planning, and learning for loco-manipulation necessitate robot hardware that has high-power-density actuators, strong yet lightweight limbs, and accurate sensing. These requirements are often limitations to hardware performance, which drives hardware designers to push the boundary of humanoid physical capabilities.

The traditional approach to increase torque density is through high gear reduction [87], and the inclusion of elastic elements to be impact-resilient [309]. However, the last decade has witnessed advancements in electric motor design such as quasi-direct drive (QDD) actuators (less than 10:1 gear ratio) achieving dynamic and accurate motions [2], [259]. QDD actuators offer the advantages of being backdrivable and enabling estimation of the output torque through the motor current [107]. Torque estimation also plays a vital role in human–robot safety, with many collision detection algorithms based on this measurement [310]. The main limitations of QDD actuators remain power density, as well as thermal and magnetic operating limits [311]. For example, achieving high torque requires a high current, which in turn causes strain on power electronics and overheating. Solving this problem involves manufacturing with materials to enhance heat dissipation [311] and continuing to improve controller performance for energy efficiency [38].

C. Future Applications and Challenges of Humanoid Robots in the Society

Given a sufficient amount of time, humanoid robots will be increasingly autonomous and capable of performing general human-level tasks across a wide range of domains. The components discussed in this article—sensing, planning, control, and learning—coordinate in real time to execute complex, goal-directed behaviors. This integration enables humanoid robots to take on tedious, hazardous, and physically demanding tasks.

1) In outdoor environments, they could serve as first responders during national crises, providing medical care and emergency support; monitor controlled settings in civil and mechanical infrastructure; deliver and sort building materials at construction sites; explore subterranean or collapsed structures with harsh and unpredictable terrain; and assist in agricultural tasks such as planting crops.
2) In factories and warehouses, humanoid robots have the potential to address the growing demand for flexible manufacturing by performing tasks such as delivery, assembly, and inspection, helping to offset the impact of a declining industrial workforce.
3) In human-centered settings such as households and hospitals, they have the potential to carry laundry, load dishwashers, and move patients between beds, supporting the needs of an aging population and improving their quality of life.

However, significant challenges remain in integrating humanoid robots into all aspects of society. To be fully accepted, these robots must be physically and mentally reliable, highlighting the core technical challenges of achieving robust and safe mobility and manipulation. Their intentions must also be well predicted and communicated with humans, which calls for advancements in effective human–robot interaction. Beyond the technical hurdles, economic and ethical concerns also pose major obstacles. Humanoid robots are still far from being commercially viable due to high costs, and the absence of legal and regulatory policies further hinders their appearance in our daily lives.

X. CONCLUSION

Humanoid robotics is advancing at an unprecedented pace, as seen in recent groundbreaking innovations from both industry and academia. In this survey, we go over foundational and state-of-the-art methods for humanoid loco-manipulation, from both model-based and learning-based perspectives. Although humanoid robots still face significant technical challenges, their agility, safety, reliability, and versatility have improved significantly. More opportunities are emerging through the exploration of new paradigms. Physically, advanced observers (vision and whole-body tactile sensing) and estimators are emerging for contact-rich whole-body loco-manipulation. Cognitively, FMs grounded on humanoid robots have great potential to unlock the ability of open-world understanding and the development of generalized intelligent agents. More importantly, an integration of FMs with planning and control policies presents promising capabilities. In the foreseeable future, the cost of humanoid robots will continue to decrease, making them more accessible; their physical capability (hardware intelligence) and cognitive intelligence will significantly advance. We look forward to humanoid robots that are responsive, purposeful, and fully capable of human-like loco-manipulation tasks in the upcoming decade.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

ACKNOWLEDGMENT

Authors' Affiliations

Zhaoyuan Gu and Ye Zhao are with the Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, GA 30332-0002 USA (e-mail: zgu78@gatech.edu; yezhao@gatech.edu).

Junheng Li and Quan Nguyen are with the Department of Aerospace and Mechanical Engineering, the University of Southern California, Los Angeles, CA 90007 USA (e-mail: junhengl@usc.edu; quann@usc.edu).

Wenlan Shen and Gordon Cheng are with the School of Computation, Information, and Technology, Technische Universität München, 80333 Munich, Germany (e-mail: wenlan.shen@tum.de; gordon@tum.de).

Wenhao Yu is with Google DeepMind, Mountain View, CA 94043 USA (e-mail: magicmelon@google.com).

Zhaoming Xie is with Robotics AI Institute, Cambridge, MA 02142 USA (e-mail: zxie@theaiinstitute.com).

Stephen McCrory and Robert Griffin are with the Institute for Human and Machine Cognition, Pensacola, FL 32502 USA (e-mail: smccrory@ihmc.org; rgriffin@ihmc.org).

Xianyi Cheng is with the Department of Mechanical Engineering and Materials Science, Duke University, Durham, NC 27708-0187 USA (e-mail: xianyi.cheng@duke.edu).

Abdulaziz Shamsah is with the College of Engineering and Petroleum, Kuwait University, Safat 13060, Kuwait (e-mail: Abdulaziz.shamsah@ku.edu.kw).

C. Karen Liu is with Computer Science Department, Stanford University, Stanford, CA 94305-6104 USA (e-mail: karenliu@cs.stanford.edu).

Abderrahmane Kheddar is with AIST Tsukuba Headquarters and Information Technology Collaborative Research Center (AIST Tsukuba Central 1), Tsukuba 305-8560, Japan (e-mail: kheddar@lirmm.fr).

Xue Bin Peng is with Computer Science Department, Simon Fraser University, Burnaby, BC V5A 1S6 USA (e-mail: xbpeng@sfu.ca).

Yuke Zhu is with the Department of Computer Science, The University of Texas at Austin, Austin, TX 78712-1139 USA (e-mail: yukez@cs.utexas.edu).

Guanya Shi is with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA 15213-3815 USA (e-mail: guanyas@andrew.cmu.edu).

Huijun Gao is with the Research Institute of Intelligent Control and Systems, Harbin Institute of Technology, Harbin 150001, China (e-mail: hjgao@hit.edu.cn).

REFERENCES

[1] H. Qiao, "Enhancing locomotion abilities of humanoid robots using universal gait templates for reinforcement learning," *IEEE/ASME Trans. Mechatronics*, vol. 30, no. 6, pp. 7242–7252, 2025, doi: 10.1109/TMECH.2025.3579364.

[2] J. Li, J. Ma, O. Kolt, M. Shah, and Q. Nguyen, "Dynamic loco-manipulation on hector: Humanoid for enhanced control and open-source research," 2023, *arXiv:2312.11868*.

[3] B. Henze, M. A. Roa, and C. Ott, "Passivity-based whole-body balancing for torque-controlled humanoid robots in multi-contact scenarios," *Int. J. Robot. Res.*, vol. 35, no. 12, pp. 1522–1543, 2016.

[4] N. Figueroa, S. Faraji, M. Koptev, and A. Billard, "A dynamical system approach for adaptive grasping, navigation and co-manipulation with humanoid robots," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 7676–7682.

[5] D. Calvert et al., "A behavior architecture for fast humanoid robot door traversals," in *Proc. Robot. Auton. Syst.*, vol. 195, 2026, p. 105217, doi: 10.1016/j.robot.2025.105217.

[6] C. Khazoom, S. Hong, M. Chignoli, E. Stanger-Jones, and S. Kim, "Tailoring solution accuracy for fast whole-body model predictive control of legged robots," *IEEE Robot. Autom. Lett.*, vol. 9, no. 12, pp. 11074–11081, Dec. 2024.

[7] A. Goswami and P. Vadakkepat, *Humanoid Robotics: A Reference*, 1st ed. Berlin, Germany: Springer, 2018.

[8] L. Zhao, Z. Yu, L. Han, X. Chen, X. Qiu, and Q. Huang, "Compliant motion control of wheel-legged humanoid robot on rough terrains," *IEEE/ASME Trans. Mechatron.*, vol. 29, no. 3, pp. 1949–1959, Jun. 2024.

[9] P. M. Wensing, M. Posa, Y. Hu, A. Escande, N. Mansard, and A. D. Prete, "Optimization-based control for dynamic legged robots," *IEEE Trans. Robot.*, vol. 40, pp. 43–63, 2024.

[10] R. Firoozi et al., "Foundation models in robotics: Applications, challenges, and the future," *Int. J. Robot. Res.*, vol. 44 2024, Art. no. 02783649241281508.

[11] Y. Hu et al., "Toward general-purpose robots via foundation models: A survey and meta-analysis," *CoRR*, vol. 2312.08782, pp. 1–42, 2023. [Online]. Available: https://doi.org/10.48550/arXiv.2312.08782

[12] R. S. Dahiya, G. Metta, M. Valle, and G. Sandini, "Tactile sensing—from humans to humanoids," *IEEE Trans. Robot.*, vol. 26, no. 1, pp. 1–20, Feb. 2010.

[13] M. M. Sotaro Katayama and Y. Tazaki, "Model predictive control of legged and humanoid robots: Models and algorithms," *Adv. Robot.*, vol. 37, no. 5, pp. 298–315, 2023.

[14] L. Bao, J. Humphreys, T. Peng, and C. Zhou, "Deep reinforcement learning for bipedal locomotion: A brief survey," *Artif. Intell. Rev.*, vol. 59, 2025, doi: 10.1007/s10462-025-11451-z.

[15] S. Ha, J. Lee, M. van de Panne, Z. Xie, W. Yu, and M. Khadiv, "Learning-based legged locomotion: State of the art and future perspectives," *Int. J. Robot. Res.*, vol. 44, no. 8, pp. 1396–1427, 2025, doi: 10.1177/02783649241312698.

[16] A. Gams, T. Petrič, B. Nemec, and A. Ude, "Manipulation learning on humanoid robots," *Curr. Robot. Rep.*, vol. 3, no. 3, pp. 97–109, 2022.

[17] J. Parra-Moreno, D. Yanguas-Rojas, and E. Mojica-Nava, "Imitation methods for bipedal locomotion of humanoid robots: A survey," in *Proc. IEEE Colombian Conf. Autom. Control*, 2023, pp. 1–6.

[18] Y. Tong, H. Liu, and Z. Zhang, "Advancements in humanoid robots: A comprehensive review and future prospects," *IEEE/CAA J. Automatica Sinica*, vol. 11, no. 2, pp. 301–328, Feb. 2024.

[19] P.-B. Wieber, R. Tedrake, and S. Kuindersma, "Modeling and control of legged robots," in *Handbook of Robotics*. Berlin, Germany: Springer, 2016, pp. 1203–1234.

[20] J. Carpentier and P.-B. Wieber, "Recent progress in legged robots locomotion control," *Curr. Robot. Rep.*, vol. 2, no. 3, pp. 231–238, 2021.

[21] M. S. Khan and R. K. Mandava, "A review on gait generation of the biped robot on various terrains," *Robotica*, vol. 41, no. 6, pp. 1888–1930, 2023.

[22] E. R. Westervelt, J. W. Grizzle, C. Chevallereau, J. H. Choi, and B. Morris, *Feedback Control of Dynamic Bipedal Robot Locomotion*. Boca Raton, FL, USA: CRC Press, 2018.

[23] S. Kajita, H. Hirukawa, K. Harada, and K. Yokoi, *Introduction to Humanoid Robotics*. Berlin, Germany: Springer, 2014.

[24] T. McGeer, "Passive walking with knees," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1990, pp. 1640–1645.

[25] S. Collins, A. Ruina, R. Tedrake, and M. Wisse, "Efficient bipedal robots based on passive-dynamic walkers," *Science*, vol. 307, no. 5712, pp. 1082–1085, 2005.

[26] S. Kajita et al., "Biped walking pattern generation by using preview control of zero-moment point," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2003, vol. 2, pp. 1620–1626.

[27] J. Englsberger, A. Dietrich, G. Mesesan, G. Garofalo, C. Ott, and A. Albu-Schäffer, "MPTC-modular passive tracking controller for stack of tasks based control frameworks," in *Proc. Robot.: Sci. Syst.*, , vol. 16, Jul. 2020, pp. 1–9, doi: 10.15607/RSS.2020.XVI.077.

[28] D. J. Agravante, A. Cherubini, A. Sherikov, P.-B. Wieber, and A. Kheddar, "Human-humanoid collaborative carrying," *IEEE Trans. Robot.*, vol. 35, no. 4, pp. 833–846, Aug. 2019.

[29] J. Morimoto and C. G. Atkeson, "Learning biped locomotion," *IEEE Robot. Autom. Mag.*, vol. 14, no. 2, pp. 41–51, Jun. 2007.

[30] J. Nakanishi, J. Morimoto, G. Endo, G. Cheng, S. Schaal, and M. Kawato, "Learning from demonstration and adaptation of biped locomotion," *Robot. Auton. Syst.*, vol. 47, no. 2, pp. 79–91, 2004.

[31] R. Calandra, A. Seyfarth, J. Peters, and M. P. Deisenroth, "Bayesian optimization for learning gaits under uncertainty: An experimental comparison on a dynamic bipedal walker," *Ann. Math. Artif. Intell.*, vol. 76, no. 1–2, pp. 5–23, 2015.

[32] D. Crowley, J. Dao, H. Duan, K. Green, J. Hurst, and A. Fern, "Optimizing bipedal locomotion for the 100 m dash with comparison to human running," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 12205–12211.

[33] Z. Li, X. B. Peng, P. Abbeel, S. Levine, G. Berseth, and K. Sreenath, "Reinforcement learning for versatile, dynamic, and robust bipedal locomotion control," *Int. J. Robot. Res.*, vol. 44, no. 5, 2024, Art. no. 02783649241285161.

[34] J. Siekmann, K. Green, J. Warila, A. Fern, and J. Hurst, "Blind bipedal stair traversal via sim-to-real reinforcement learning," in *Proc. Robot.: Sci. Syst.*, vol. 17, 2022, pp. 1–9.

[35] Z. Zhuang, S. Yao, and H. Zhao, "Humanoid parkour learning," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–17. [Online]. Available: https://openreview.net/forum?id=fs7ia3FqUM

[36] J. Dao, K. Green, H. Duan, A. Fern, and J. Hurst, "Sim-to-real learning for bipedal locomotion under unsensed dynamic loads," in *Proc. Int. Conf. Robot. Autom.*, 2022, pp. 10449–10455.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

[37] I. Radosavovic, T. Xiao, B. Zhang, T. Darrell, J. Malik, and K. Sreenath, "Real-world humanoid locomotion with reinforcement learning," *Sci. Robot.*, vol. 9, no. 89, 2024, Art. no. eadi9579.
[38] N. Kashiri et al., "An overview on principles for energy efficient robot locomotion," *Front. Robot. AI*, vol. 5, 2018, Art. no. 129.
[39] J. Reher, E. A. Cousineau, A. Hereid, C. M. Hubicki, and A. D. Ames, "Realizing dynamic and efficient bipedal locomotion on the humanoid robot durus," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 1794–1801.
[40] F. Heremans, S. Vijayakumar, M. Bouri, B. Dehez, and R. Ronsse, "Bio-inspired design and validation of the efficient lockable spring ankle (ELSA) prosthesis," in *Proc. IEEE Int. Conf. Rehabil. Robot.*, Jun. 2019, vol. 2019, pp. 411–416.
[41] B. Xia, B. Li, J. Lee, M. Scutari, and B. Chen, "The duke humanoid: Design and control for energy efficient bipedal locomotion using passive dynamics," *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2025, pp. 6579–6586, doi: 10.1109/IROS60139.2025.11246828.
[42] C. Zhang, J. Gao, Z. Chen, S. Zhong, and H. Qiao, "Fall analysis and prediction for humanoids," *Robot. Auton. Syst.*, vol. 190, 2025, Art. no. 104995.
[43] Z. Gu et al., "Robust-locomotion-by-logic: Perturbation-resilient bipedal locomotion via signal temporal logic guided model predictive control," *IEEE Trans. Robot.*, vol. 41, pp. 4300–4321, 2025, doi: 10.1109/TRO.2025.3582820.
[44] O. Stasse and L. Righetti, *Whole-Body Manipulation*. Berlin, Germany: Springer, 2020, pp. 1–9.
[45] J. K. Salisbury, "Whole-arm manipulation," in *Proc. Int. Symp. Robot. Res.*, 1987, pp. 183–189.
[46] W. T. Townsend and J. K. Salisbury, "Mechanical design for whole-arm manipulation," in *Robots and Biological Systems: Towards a New Bionics*. Berlin, Germany: Springer, 1993, pp. 153–164.
[47] A. Goncalves, N. Kuppuswamy, A. Beaulieu, A. Uttamchandani, K. M. Tsui, and A. Alspach, "Punyo-1: Soft tactile-sensing upper-body robot for large object manipulation and physical human interaction," in *Proc. IEEE Int. Conf. Soft Robot.*, 2022, pp. 844–851.
[48] C. Smith et al., "Dual arm manipulation—A survey," *Robot. Auton. Syst.*, vol. 60, no. 10, pp. 1340–1353, 2012.
[49] K. Hsiao and T. Lozano-Perez, "Imitation learning of whole-body grasps," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2006, pp. 5657–5662.
[50] M. Zhang, J. Barreiros, and A. O. Onol, "Plan-guided reinforcement learning for whole-body manipulation," in *Proc. IROS Workshop Leveraging Models Contact-Rich Manipulation*, 2023, pp. 351–360. [Online]. Available: https://openreview.net/forum?id=Ju2M8XN2jA
[51] G. Cheng et al., "A comprehensive realization of robot skin: Sensors, sensing, control, and applications," *Proc. IEEE*, vol. 107, no. 10, pp. 2034–2051, Oct. 2019.
[52] M. Florek-Jasińska, T. Wimböck, and C. Ott, "Humanoid compliant whole arm dexterous manipulation: Control design and experiments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 1616–1621.
[53] M. Murooka, S. Nozawa, Y. Kakiuchi, K. Okada, and M. Inaba, "Whole-body pushing manipulation with contact posture planning of large and heavy object for humanoid robot," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 5682–5689.
[54] T. Zhang et al., "Deep imitation learning for complex manipulation tasks from virtual reality teleoperation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 5628–5635.
[55] D. Rakita, B. Mutlu, M. Gleicher, and L. M. Hiatt, "Shared control–based bimanual robot manipulation," *Sci. Robot.*, vol. 4, no. 30, 2019, Art. no. eaaw0955.
[56] K. Chappellet, M. Murooka, G. Caron, F. Kanehiro, and A. Kheddar, "Humanoid loco-manipulations using combined fast dense 3d tracking and slam with wide-angle depth-images," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3691–3704, Jul. 2024.
[57] M. Seo et al., "Deep imitation learning for humanoid loco-manipulation through human teleoperation," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2023, pp. 1–8.
[58] Z. Kappassov, J.-A. Corrales, and V. Perdereau, "Tactile sensing in dexterous robot hands," *Robot. Auton. Syst.*, vol. 74, pp. 195–220, 2015.
[59] Y. Al-Handarish et al., "A survey of tactile-sensing systems and their applications in biomedical engineering," *Adv. Mater. Sci. Eng.*, vol. 2020, 2020, Art. no. 4047937.
[60] Q. Li, C. Schürmann, R. Haschke, and H. J. Ritter, "A control framework for tactile servoing," in *Proc. Robot.: Sci. Syst.*, vol. 9, 2013, pp. 1–8.
[61] F. Veiga, B. Edin, and J. Peters, "Grip stabilization through independent finger tactile feedback control," *Sensors*, vol. 20, no. 6, 2020, Art. no. 1748.
[62] X. Song, H. Liu, K. Althoefer, T. Nanayakkara, and L. D. Seneviratne, "Efficient break-away friction ratio and slip prediction based on haptic surface exploration," *IEEE Trans. Robot.*, vol. 30, no. 1, pp. 203–219, Feb. 2014.
[63] E. Y. P. Mittendorfer and G. Cheng, "Realizing whole-body tactile interactions with a self-organizing, multi-modal artificial skin on a humanoid robot," *Adv. Robot.*, vol. 29, no. 1, pp. 51–67, 2015.
[64] N. Sommer, M. Li, and A. Billard, "Bimanual compliant tactile exploration for grasping unknown objects," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 6400–6407.
[65] F. R. Hogan, J. Ballester, S. Dong, and A. Rodriguez, "Tactile dexterity: Manipulation primitives with tactile feedback," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 8863–8869.
[66] M. Kaboli and G. Cheng, "Robust tactile descriptors for discriminating objects from textural properties via artificial robotic skin," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 985–1003, Aug. 2018.
[67] S. Tian et al., "Manipulation by feel: Touch-based control with deep predictive models," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 818–824.
[68] M. Lambeta et al., "Digit: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation," *Robot. Autom. Lett.*, vol. 5, no. 3, pp. 3838–3845, 2020.
[69] H. Van Hoof, T. Hermans, G. Neumann, and J. Peters, "Learning robot in-hand manipulation with tactile features," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2015, pp. 121–127.
[70] A. Melnik, L. Lach, M. Plappert, T. Korthals, R. Haschke, and H. Ritter, "Using tactile sensing to improve the sample efficiency and performance of deep deterministic policy gradients for simulated in-hand manipulation tasks," *Front. Robot. AI*, vol. 8, 2021, Art. no. 538773.
[71] T. Lin et al., "Learning visuotactile skills with two multifingered hands," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2025, pp. 5637–5643, doi: 10.1109/ICRA55743.2025.11128180.
[72] L. Fu et al., "A touch, vision, and language dataset for multimodal alignment," in *Proc. 41st Int. Conf. Mach. Learn.*, vol. 235, 2024, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=tFEOOH9eH0
[73] S. Yu, L. Kelvin, A. Xiao, J. Duan, and H. Soh, "Octopi: Object property reasoning with large tactile-language models," in *Proc. Robot.: Sci. Syst.*, vol. 20, 2024, pp. 1–17, doi: 10.15607/RSS.2024.XX.066.
[74] Y. Chebotar, O. Kroemer, and J. Peters, "Learning robot tactile sensing for object manipulation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 3368–3375.
[75] H. Van Hoof, N. Chen, M. Karl, P. van der Smagt, and J. Peters, "Stable reinforcement learning with autoencoders for tactile and visual data," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 3928–3934.
[76] S. Wang, M. Lambeta, P.-W. Chou, and R. Calandra, "Tacto: A fast, flexible, and open-source simulator for high-resolution vision-based tactile sensors," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 3930–3937, Apr. 2022.
[77] Y. Lin, J. Lloyd, A. Church, and N. F. Lepora, "Tactile gym 2.0: Sim-to-real deep reinforcement learning for comparing low-cost high-resolution robot touch," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 10754–10761, Oct. 2022.
[78] I. Akinola, J. Xu, J. Carius, D. Fox, and Y. Narang, "TacSL: A library for visuotactile sensor simulation and learning," *IEEE Trans. Robot.*, vol. 41, pp. 2645–2661, 2025.
[79] J. R. Guadarrama-Olvera, F. Bergner, E. Dean, and G. Cheng, "Enhancing biped locomotion on unknown terrain using tactile feedback," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2018, pp. 1–9.
[80] X. Guo et al., "Soft foot sensor design and terrain classification for dynamic legged locomotion," in *Proc. IEEE Int. Conf. Soft Robot.*, 2020, pp. 550–557.
[81] K. Suwanratchatamanee, M. Matsumoto, and S. Hashimoto, "A simple tactile sensing foot for humanoid robot and active ground slope recognition," in *Proc. IEEE Int. Conf. Mechatron.*, 2009, pp. 1–6.
[82] F. Veiga, R. Akrour, and J. Peters, "Hierarchical tactile-based control decomposition of dexterous in-hand manipulation tasks," *Front. Robot. AI*, vol. 7, 2020, Art. no. 521448.
[83] P. Maiolino, M. Maggiali, G. Cannata, G. Metta, and L. Natale, "A flexible and robust large scale capacitive tactile system for robots," *IEEE Sensors J.*, vol. 13, no. 10, pp. 3910–3917, Oct. 2013.
[84] E. Dean-Leon, J. R. Guadarrama-Olvera, F. Bergner, and G. Cheng, "Whole-body active compliance control for humanoid robots with robot skin," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 5404–5410.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

[85] T. Kobayashi, E. Dean-Leon, J. R. Guadarrama-Olvera, F. Bergner, and G. Cheng, "Whole-body multicontact haptic human–humanoid interaction based on leader–follower switching: A robot dance of the "box step"," *Adv. Intell. Syst.*, vol. 4, no. 2, 2022, Art. no. 2100038.

[86] W. Yuan, S. Dong, and E. H. Adelson, "Gelsight: High-resolution robot tactile sensors for estimating geometry and force," *Sensors*, vol. 17, no. 12, 2017, Art. no. 2762.

[87] F. Nori, S. Traversaro, J. Eljaik, F. Romano, A. Del Prete, and D. Pucci, "iCUB whole-body control through force regulation on rigid non-coplanar contacts," *Front. Robot. AI*, vol. 2, pp. 1–18, 2015.

[88] A. Jain, M. D. Killpack, A. Edsinger, and C. C. Kemp, "Reaching in clutter with whole-arm tactile sensing," *Int. J. Robot. Res.*, vol. 32, no. 4, pp. 458–482, 2013.

[89] A. Kheddar et al., "Humanoid robots in aircraft manufacturing: The airbus use cases," *IEEE Robot. Autom. Mag.*, vol. 26, no. 4, pp. 30–45, Dec. 2019.

[90] Y. Ohmura and Y. Kuniyoshi, "Humanoid robot which can lift a 30 kg box by whole body contact and tactile feedback," in *Proc. Int. Conf. Intell. Robots Syst.*, 2007, pp. 1136–1141.

[91] L. Natale and G. Cannata, "Tactile sensing," in *Humanoid Robotics: A Reference*. Berlin, Germany: Springer, 2017, pp. 110–1.

[92] M. Posa, C. Cantu, and R. Tedrake, "A direct method for trajectory optimization of rigid bodies through contact," *Int. J. Robot. Res.*, vol. 33, no. 1, pp. 69–81, 2014.

[93] I. Mordatch, E. Todorov, and Z. Popović, "Discovery of complex behaviors through contact-invariant optimization," *ACM Trans. Graph.*, vol. 31, no. 4, pp. 1–8, 2012.

[94] I. Chatzinikolaidis, Y. You, and Z. Li, "Contact-implicit trajectory optimization using an analytically solvable contact model for locomotion on variable ground," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 6357–6364, Oct. 2020.

[95] Z. Manchester and S. Kuindersma, "Variational contact-implicit trajectory optimization," in *Proc. Robot. Research: Int. Symp.*, 2020, pp. 985–1000.

[96] M. Posa, S. Kuindersma, and R. Tedrake, "Optimization and stabilization of trajectories for constrained dynamical systems," in *Proc. Int. Conf. Robot. Autom.*, 2016, pp. 1366–1373.

[97] S. Le Cleac'h et al., "Fast contact-implicit model predictive control," *Trans. Robot.*, vol. 40, pp. 1617–1629, 2024.

[98] N. J. Kong, C. Li, G. Council, and A. M. Johnson, "Hybrid iLQR model predictive control for contact implicit stabilization on legged robots," *IEEE Trans. Robot.*, vol. 39, no. 6, pp. 4712–4727, Dec. 2023.

[99] A. Aydinoglu, A. Wei, W.-C. Huang, and M. Posa, "Consensus complementarity control for multi-contact mpc," *IEEE Trans. Robot.*, vol. 40, pp. 3879–3896, 2024.

[100] V. Kurtz, A. Castro, A. Ö. Önol, and H. Lin, "Inverse dynamics trajectory optimization for contact-implicit model predictive control," *Int. J. Robot. Res.*, vol. 45, no. 1, pp. 23–40, 2026, doi: 10.1177/02783649251344635.

[101] G. B. Margolis et al., "Learning to jump from pixels," in *Proc. Conf. Robot Learn.*, vol. 164, 2022, pp. 1025–1034.

[102] V. Tsounis, M. Alge, J. Lee, F. Farshidian, and M. Hutter, "Deepgait: Planning and control of quadrupedal gaits using deep reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3699–3706, Apr. 2020.

[103] S. Xu, Z. Li, Y.-X. Wang, and L.-Y. Gui, "Interdiff: Generating 3d human-object interactions with physics-informed diffusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 14928–14940.

[104] J. Li, J. Wu, and C. K. Liu, "Object motion guided human motion synthesis," *ACM Trans. Graph.*, vol. 42, no. 6, pp. 1–11, 2023.

[105] S. Bahl, R. Mendonca, L. Chen, U. Jain, and D. Pathak, "Affordances from human videos as a versatile representation for robotics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 13778–13790.

[106] K. Bouyarmane, S. Caron, A. Escande, and A. Kheddar, "Multi-contact motion planning and control," in *Humanoid Robotics: A Reference*. Berlin, Germany: Springer, 2019, pp. 1763–1804.

[107] M. Chignoli, D. Kim, E. Stanger-Jones, and S. Kim, "The mit humanoid robot: Design, motion planning, and control for acrobatic behaviors," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2021, pp. 1–8.

[108] G. Romualdi, S. Dafarra, G. L'Erario, I. Sorrentino, S. Traversaro, and D. Pucci, "Online non-linear centroidal mpc for humanoid robot locomotion with step adjustment," in *Proc. Int. Conf. Robot. Autom.*, 2022, pp. 10412–10419.

[109] M. Elobaid, G. Romualdi, G. Nava, L. Rapetti, H. A. O. Mohamed, and D. Pucci, "Online non-linear centroidal mpc for humanoid robots payload carrying with contact-stable force parametrization," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 12233–12239.

[110] B. Henze, C. Ott, and M. A. Roa, "Posture and balance control for humanoid robots in multi-contact scenarios based on model predictive control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 3253–3258.

[111] S. H. Bang et al., "Rpc: A modular framework for robot planning, control, and deployment," in *Proc. IEEE/SICE Int. Symp. Syst. Integration*, 2025, pp. 1142–1148.

[112] L. Penco, N. Scianca, V. Modugno, L. Lanari, G. Oriolo, and S. Ivaldi, "A multimode teleoperation framework for humanoid loco-manipulation: An application for the iCUB robot," *IEEE Robot. Autom. Mag.*, vol. 26, no. 4, pp. 73–82, Dec. 2019.

[113] S. Kuindersma et al., "Optimization-based locomotion planning, estimation, and control design for the atlas humanoid robot," *Auton. Robots*, vol. 40, no. 3, pp. 429–455, 2016.

[114] H. Dai, A. Valenzuela, and R. Tedrake, "Whole-body motion planning with centroidal dynamics and full Kinematics," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2014, pp. 295–302.

[115] M. P. Polverini, A. Laurenzi, E. M. Hoffman, F. Ruscelli, and N. G. Tsagarakis, "Multi-contact heavy object pushing with a centaur-type humanoid robot: Planning and control for a real demonstrator," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 859–866, Apr. 2020.

[116] E. Dantec et al., "Whole body model predictive control with a memory of motion: Experiments on a torque-controlled talos," in *Proc. Int. Conf. Robot. Autom.*, 2021, pp. 8202–8208.

[117] J. Li and Q. Nguyen, "Force-and-moment-based model predictive control for achieving highly dynamic locomotion on bipedal robots," in *Proc. IEEE Conf. Decis. Control*, 2021, pp. 1024–1030.

[118] H. Audren, J. Vaillant, A. Kheddar, A. Escande, K. Kaneko, and E. Yoshida, "Model preview control in multi-contact motion-application to a humanoid robot," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 4030–4035.

[119] D. E. Orin, A. Goswami, and S.-H. Lee, "Centroidal dynamics of a humanoid robot," *Auton. Robots*, vol. 35, pp. 161–176, 2013.

[120] H. Li, R. J. Frei, and P. M. Wensing, "Model hierarchy predictive control of robotic systems," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3373–3380, Apr. 2021.

[121] J. Wang, S. Kim, S. Vijayakumar, and S. Tonneau, "Multi-fidelity receding horizon planning for multi-contact locomotion," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2021, pp. 53–60.

[122] J. Li, Z. Le, J. Ma, and Q. Nguyen, "Adapting gait frequency for posture-regulating humanoid push-recovery via hierarchical model predictive control," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2024, pp. 12330–12336, doi: 10.1109/ICRA55743.2025.11127608.

[123] R. Budhiraja, J. Carpentier, and N. Mansard, "Dynamics consensus between centroidal and whole-body models for locomotion of legged robots," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 6727–6733.

[124] A. Herzog, S. Schaal, and L. Righetti, "Structured contact force optimization for Kino-dynamic motion generation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst*., 2016, pp. 2703–2710.

[125] D. Mayne, "A second-order gradient method for determining optimal trajectories of non-linear discrete-time systems," *Int. J. Control*, vol. 3, no. 1, pp. 85–95, 1966.

[126] Y. Tassa, T. Erez, and E. Todorov, "Synthesis and stabilization of complex behaviors through online trajectory optimization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2012, pp. 4906–4913.

[127] L. B. Rall, *Automatic Differentiation: Techniques and Applications*. Berlin, Germany: Springer, 1981.

[128] C. Rösmann, M. Krämer, A. Makarow, F. Hoffmann, and T. Bertram, "Exploiting sparse structures in nonlinear model predictive control with hypergraphs," in *Proc. IEEE/ASME Int. Conf. Adv. Intell. Mechatron.*, 2018, pp. 1332–1337.

[129] L. Vanroye, A. Sathya, J. De Schutter, and W. Decré, "Fatrop: A fast constrained optimal control problem solver for robot trajectory optimization and control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2023, pp. 10036–10043.

[130] L. Callens, J. Gillis, W. Decré, and J. Swevers, "Adaptivenlp: A framework for efficient online adaptability in NLP structures for optimal control problems," in *Proc. Eur. Control Conf.*, 2024, pp. 1358–1365.

[131] R. B. Wilson, "A simplicial method for convex programming," Ph.D. dissertation, Graduate Sch. Bus. Admin., Harvard Univ., 1963.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

[132] A. Zhakatayev, B. Rakhim, O. Adiyatov, A. Baimyshev, and H. A. Varol, "Successive linearization based model predictive control of variable stiffness actuated robots," in *Proc. IEEE Int. Conf. Adv. Intell. Mechatron.*, 2017, pp. 1774–1779.

[133] J. L. Jerez, E. C. Kerrigan, and G. A. Constantinides, "A condensed and sparse QP formulation for predictive control," in *Proc. 50th IEEE Conf. Decis. Control Eur. Control*, 2011, pp. 5217–5222.

[134] A. Bishop, J. Zhang, S. Gurumurthy, K. Tracy, and Z. Manchester, "Relu-qp: A gpu-accelerated quadratic programming solver for model-predictive control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 13285–13292.

[135] X. Da and J. Grizzle, "Combining trajectory optimization, supervised machine learning, and model structure for mitigating the curse of dimensionality in the control of bipedal robots," *Int. J. Robot. Res.*, vol. 38, no. 9, pp. 1063–1097, 2019.

[136] Y. Ding, A. Pandala, and H.-W. Park, "Real-time model predictive control for versatile dynamic motions in quadrupedal robots," in *Proc. Int. Conf. Robot. Autom.*, 2019, pp. 8484–8490.

[137] P. Ferrari et al., "Multi-contact planning and control for humanoid robots: Design and validation of a complete framework," *Robot. Auton. Syst.*, vol. 166, 2023, Art. no. 104448.

[138] J. Li and Q. Nguyen, "Kinodynamic pose optimization for humanoid loco-manipulation," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2023, pp. 1–8.

[139] J. Li and Q. Nguyen, "Multi-contact mpc for dynamic loco-manipulation on humanoid robots," in *Proc. Amer. Control Conf.*, 2023, pp. 1215–1220.

[140] S.-C. Hsu, X. Xu, and A. D. Ames, "Control barrier function based quadratic programs with application to bipedal robotic walking," in *Proc. Amer. Control Conf.*, 2015, pp. 4542–4548.

[141] S. Yu, N. Perera, D. Marew, and D. Kim, "Learning generic and dynamic locomotion of humanoids across discrete terrains," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2024, pp. 1048–1055.

[142] K. Darvish et al., "Teleoperation of humanoid robots: A survey," *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 1706–1727, Jun. 2023.

[143] P. Marion et al., "Director: A user interface designed for robot operation with shared autonomy," *Proc. DARPA Robot. Challenge Finals: Humanoid Robots Rescue*, pp. 237–270, 2018.

[144] L. Penco et al., "Robust real-time whole-body motion retargeting from human to humanoid," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2018, pp. 425–432.

[145] Q. Rouxel, K. Yuan, R. Wen, and Z. Li, "Multicontact motion retargeting using whole-body optimization of full kinematics and sequential force equilibrium," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 5, pp. 4188–4198, Oct. 2022.

[146] S. J. Jorgensen, M. Wonsick, M. Paterson, A. Watson, I. Chase, and J. S. Mehling, "Cockpit interface for locomotion and manipulation control of the nasa valkyrie humanoid in virtual reality (VR)," Nat, Aeronaut, Space Admin., White Paper, 2022.

[147] D. Wallace, Y. H. He, J. C. Vaz, L. Georgescu, and P. Y. Oh, "Multimodal teleoperation of heterogeneous robots within a construction environment," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 2698–2705.

[148] L. Sentis, "Synthesis and control of whole-body behaviors in humanoid systems," Ph.D. dissertation, Dept. Elect. Eng., Stanford Univ., Stanford, CA, USA, 2007.

[149] M. Mistry, J. Buchli, and S. Schaal, "Inverse dynamics control of floating base systems using orthogonal decomposition," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2010, pp. 3406–3412.

[150] L. Righetti, J. Buchli, M. Mistry, and S. Schaal, "Inverse dynamics control of floating-base robots with external constraints: A unified view," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2011, pp. 1085–1090.

[151] J. Nakanishi, R. Cory, M. Mistry, J. Peters, and S. Schaal, "Operational space control: A theoretical and empirical comparison," *Int. J. Robot. Res.*, vol. 27, no. 6, pp. 737–757, 2008.

[152] N. Mansard, O. Khatib, and A. Kheddar, "A unified approach to integrate unilateral constraints in the stack of tasks," *IEEE Trans. Robot.*, vol. 25, no. 3, pp. 670–685, Jun. 2009.

[153] F. Keith, P.-B. Wieber, N. Mansard, and A. Kheddar, "Analysis of the discontinuities in prioritized tasks-space control under discreet task scheduling operations," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 3887–3892.

[154] A. Rocchi, E. M. Hoffman, D. G. Caldwell, and N. G. Tsagarakis, "OpenSoT: A whole-body control library for the compliant humanoid robot COMAN," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 6248–6253.

[155] A. Escande, N. Mansard, and P.-B. Wieber, "Hierarchical quadratic programming: Fast online humanoid-robot motion generation," *Int. J. Robot. Res.*, vol. 33, no. 7, pp. 1006–1028, 2014.

[156] K. Bouyarmane and A. Kheddar, "Using a multi-objective controller to synthesize simulated humanoid robot motion with changing contact configurations," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 4414–4419.

[157] T. Koolen et al., "Design of a momentum-based control framework and application to the humanoid robot Atlas," *Int. J. Humanoid Robot.*, vol. 13, no. 01, 2016, Art. no. 1650007.

[158] M. A. Hopkins, D. W. Hong, and A. Leonessa, "Compliant locomotion using whole-body control and divergent component of motion tracking," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 5726–5733.

[159] K. Pfeiffer, A. Escande, and A. Kheddar, "Singularity resolution in equality and inequality constrained hierarchical task-space control by adaptive nonlinear least squares," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3630–3637, Oct. 2018.

[160] K. Bouyarmane and A. Kheddar, "On weight-prioritized multitask control of humanoid robots," *IEEE Trans. Autom. Control*, vol. 63, no. 6, pp. 1632–1647, Jun. 2018.

[161] M. Djeha, P. Gergondet, and A. Kheddar, "Robust task-space quadratic programming for kinematic-controlled robots," *IEEE Trans. Robot.*, vol. 39, no. 5, pp. 3857–3874, Oct. 2023.

[162] H. Hirukawa et al., "A universal stability criterion of the foot contact of legged robots—adios ZMP," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2006, pp. 1976–1983.

[163] M. Murooka et al., "Humanoid loco-manipulations pattern generation and stabilization control," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 5597–5604, Jul. 2021.

[164] L. Righetti, J. Buchli, M. Mistry, M. Kalakrishnan, and S. Schaal, "Optimal distribution of contact forces with inverse-dynamics control," *Int. J. Robot. Res.*, vol. 32, no. 3, pp. 280–298, 2013.

[165] P.-B. Wieber, *Holonomy and Nonholonomy in the Dynamics of Articulated Motion*. Berlin, Germany: Springer, 2006, pp. 411–425.

[166] K. Bouyarmane, K. Chappellet, J. Vaillant, and A. Kheddar, "Quadratic programming for multirobot and task-space force control," *IEEE Trans. Robot.*, vol. 35, no. 1, pp. 64–77, Feb. 2019.

[167] K. Otani and K. Bouyarmane, "Adaptive whole-body manipulation in human-to-humanoid multi-contact motion retargeting," in *Proc. IEEE-RAS Int. Conf. Humanoid Robot.*, 2017, pp. 446–453.

[168] M. Ji et al., "Exbody2: Advanced expressive humanoid whole-body control," in *Proc. RSS Workshop Whole-body Control Bimanual Manipulation: Appl. Humanoids Beyond*, vol. 21, 2025, pp. 1–17. [Online]. Available: https://openreview.net/forum?id=rzG7xQPSCE

[169] T. He et al., "Omnih2o: Universal and dexterous human-to-humanoid whole-body teleoperation and learning," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–25. [Online]. Available: https://openreview.net/forum?id=oL1WEZQal8

[170] Z. Fu, Q. Zhao, Q. Wu, G. Wetzstein, and C. Finn, "Humanplus: Humanoid shadowing and imitation from humans," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–17. [Online]. Available: https://openreview.net/forum?id=WnSl42M9Z4

[171] Y. Wang, N. Dehio, A. Tanguy, and A. Kheddar, "Impact-aware task-space quadratic-programming control," *Int. J. Robot. Res.*, vol. 42, no. 14, pp. 1265–1282, 2023.

[172] A. Patel, S. L. Shield, S. Kazi, A. M. Johnson, and L. T. Biegler, "Contact-implicit trajectory optimization using orthogonal collocation," *IEEE Robot. Autom. Lett.*, vol. 4, no. 2, pp. 2242–2249, Apr. 2019.

[173] J. Carius, R. Ranftl, V. Koltun, and M. Hutter, "Trajectory optimization with implicit hard contacts," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3316–3323, Oct. 2018.

[174] A. Aydinoglu and M. Posa, "Real-time multi-contact model predictive control via admm," in *Proc. Int. Conf. Robot. Autom.*, 2022, pp. 3414–3421.

[175] R. Jiang, B. He, Z. Wang, X. Cheng, H. Sang, and Y. Zhou, "Robot skill learning and the data dilemma it faces: A systematic review," *Robot. Intell. Autom.*, vol. 44, no. 2, pp. 270–286, 2024.

[176] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.

[177] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. Int. Conf. Mach. Learn.*, vol. 80, 2018, pp. 1856–1865.

[178] C. Sferrazza, D.-M. Huang, X. Lin, Y. Lee, and P. Abbeel, "Humanoid-Bench: Simulated humanoid benchmark for whole-body locomotion and

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

manipulation," in *Proc. Robot.: Sci. Syst.*, vol. 20, 2024, pp. 1–24, doi: 10.15607/RSS.2024.XX.061.
[179] V. Mnih et al., "Asynchronous methods for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1928–1937.
[180] M. Mittal et al., "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3740–3747, Jun. 2023.
[181] S. Fujimoto, H. Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 1582–1591.
[182] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
[183] Z. Xie, P. Clary, J. Dao, P. Morais, J. Hurst, and M. van de Panne, "Learning locomotion skills for cassie: Iterative design and sim-to-real," in *Proc. Conf. Robot Learn.*, vol. 100, 2020, pp. 317–329.
[184] C. Schwarke, V. Klemm, M. v. d. Boon, M. Bjelonic, and M. Hutter, "Curiosity-driven learning of joint locomotion and manipulation tasks," in *Proc. Conf. Robot Learn.*, 2023, vol. 229, pp. 2594–2610.
[185] C. Zhang, W. Xiao, T. He, and G. Shi, "Wococo: Learning whole-body humanoid control with sequential contacts," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–18. [Online]. Available: https://openreview.net/forum?id=Czs2xH9114
[186] Y. Kim et al., "Not only rewards but also constraints: Applications on legged robot locomotion," *IEEE Trans. Robot.*, vol. 40, pp. 2984–3003, 2024.
[187] T. Haarnoja et al., "Learning agile soccer skills for a bipedal robot with deep reinforcement learning," *Sci. Robot.*, vol. 9, no. 89, 2024, Art. no. eadi8022.
[188] J. Tan, Z. Xie, B. Boots, and C. K. Liu, "Simulation-based design of dynamic controllers for humanoid balancing," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 2729–2736.
[189] Y. Li, J. Li, W. Fu, and Y. Wu, "Learning agile bipedal motions on a quadrupedal robot," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 9735–9742.
[190] T. He et al., "Asap: Aligning simulation and real-world physics for learning agile humanoid whole-body skills," in *Proc. Robot.: Sci. Syst.*, vol. 21, Jun. 2025, pp. 1–18, doi: 10.15607/RSS.2025.XXI.066.
[191] K.-C. Hsu, A. Z. Ren, D. P. Nguyen, A. Majumdar, and J. F. Fisac, "Sim-to-lab-to-real: Safe reinforcement learning with shielding and generalization guarantees," *Artif. Intell.*, vol. 134, 2022, Art. no. 103811.
[192] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2012, pp. 5026–5033.
[193] J. Ibarz, J. Tan, C. Finn, M. Kalakrishnan, P. Pastor, and S. Levine, "How to train your robot with deep reinforcement learning: Lessons we have learned," *Int. J. Robot. Res.*, vol. 40, no. 4–5, pp. 698–721, 2021.
[194] X. Huang et al., "Diffuseloco: Real-time legged locomotion control with diffusion from offline datasets," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=nVJm2RdPDu
[195] F. Wu, Z. Gu, H. Wu, A. Wu, and Y. Zhao, "Infer and adapt: Bipedal locomotion reward learning from demonstrations via inverse reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 16243–16250.
[196] X. Cheng, J. Li, S. Yang, G. Yang, and X. Wang, "Open-television: Teleoperation with immersive active visual feedback," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=Yce2jeILGt
[197] Y. Wang, J. Lin, A. Zeng, Z. Luo, J. Zhang, and L. Zhang, "Physhoi: Physics-based imitation of dynamic human-object interaction," 2023, *arXiv:2312.04393*.
[198] J. Merel et al., "Learning human behaviors from motion capture by adversarial imitation," 2017, *arXiv:1707.02201*.
[199] A. Tang et al., "Humanmimic: Learning natural locomotion and transitions for humanoid robot via wasserstein adversarial imitation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 13107–13114.
[200] Q. Zhang et al., "Whole-body humanoid robot locomotion with human reference," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2024, pp. 11225–11231.
[201] X. B. Peng, A. Kanazawa, J. Malik, P. Abbeel, and S. Levine, "SFV: Reinforcement learning of physical skills from videos," *ACM Trans. On Graph.*, vol. 37, no. 6, pp. 1–14, 2018.
[202] H. Zhang, Y. Liu, and W. Zhou, "Deep adversarial imitation learning of locomotion skills from one-shot video demonstration," in *Proc. IEEE Annu. Int. Conf. CYBER Technol. Autom., Control, Intell. Syst.*, 2019, pp. 1257–1261.
[203] J. Li et al., "Okami: Teaching humanoid robots manipulation skills through single video imitation," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–19.
[204] C. Chi et al., "Diffusion policy: Visuomotor policy learning via action diffusion," *Int. J. Robot. Res.*, vol. 44, no. 10-11, pp. 1684–1704, 2025, doi: 10.1177/02783649241273668.
[205] R. Cisneros-Limón et al., "A cybernetic avatar system to embody human telepresence for connectivity, exploration, and skill transfer," *Int. J. Social Robot.*, vol. 17, no. 3, pp. 535–562, Jan. 2024, doi: 10.1007/s12369-023-01096-9.
[206] C. Lu et al., "Mobile-television: Predictive motion priors for humanoid whole-body control," in *Proc. IEEE Int. Conf. Robot. Automat.*, Atlanta, GA, USA, 2025.
[207] A. Dallard, M. Benallegue, F. Kanehiro, and A. Kheddar, "Synchronized human-humanoid motion imitation," *IEEE Robot. Autom. Lett.*, vol. 8, no. 7, pp. 4155–4162, Jul. 2023.
[208] J. Ramos and S. Kim, "Humanoid dynamic synchronization through whole-body bilateral feedback teleoperation," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 953–965, Aug. 2018.
[209] X. Xu, Y. Zhang, Y.-L. Li, L. Han, and C. Lu, "HumanVLA: Towards vision-language directed object rearrangement by physical humanoid," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, vol. 38, 2024, pp. 1–20. [Online]. Available: https://openreview.net/forum?id=pjD08dtAh0
[210] J. Juravsky, Y. Guo, S. Fidler, and X. B. Peng, "Superpadl: Scaling language-directed physics-based control with progressive supervised distillation," in *Proc. Special Int. Group Comput. Graph. Interactive Techn. Conf. Papers*. Jul. 2024, pp. 1–11.
[211] M. A. Goodrich, J. W. Crandall, and E. Barakova, "Teleoperation and beyond for assistive humanoid robots," *Rev. Hum. Factors Ergonom.*, vol. 9, no. 1, pp. 175–226, 2013.
[212] B. L. Bhatnagar, X. Xie, I. A. Petrov, C. Sminchisescu, C. Theobalt, and G. Pons-Moll, "Behave: Dataset and method for tracking human object interactions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 15935–15946.
[213] Carnegie Mellon University, "Cmu graphics lab motion capture database," 2002. [Online]. Available: http://mocap.cs.cmu.edu/
[214] Simon Fraser University and National University of Singapore, "SFU motion capture database." [Online]. Available: http://mocap.cs.sfu.ca/
[215] F. G. Harvey, M. Yurick, D. Nowrouzezahrai, and C. Pal, "Robust motion in-betweening," *ACM Trans. Graph.*, vol. 39, no. 4, pp. 60–61 2020.
[216] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, "AMASS: Archive of motion capture as surface shapes," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 5442–5451.
[217] A. Blattmann et al., "Stable video diffusion: Scaling latent video diffusion models to large datasets," *CoRR*, vol. abs/2311.15127, pp. 1–30, 2023. [Online]. Available: https://doi.org/10.48550/arXiv.2311.15127
[218] W. Zhu et al., "Human motion generation: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 2430–2449, Apr. 2024.
[219] X. Cheng, Y. Ji, J. Chen, R. Yang, G. Yang, and X. Wang, "Expressive whole-body control for humanoid robots," in *Proc. Robot.: Sci. Syst.*, Delft, Netherlands, vol. 20, Jul. 2024, pp. 1–17, doi: 10.15607/RSS.2024.XX.107.
[220] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa, "AMP: Adversarial motion priors for stylized physics-based character control," *ACM Trans. Graph.*, vol. 40, no. 4, pp. 1–20, 2021.
[221] P. Dugar, A. Shrestha, F. Yu, B. van Marum, and A. Fern, "Learning multi-modal whole-body control for real-world humanoid robots," *Proc. AAAI Symp. Ser,*, 2025, vol. 7, no. 1, pp. 650–657, doi: 10.1609/aaaiss.v7i1.36946.
[222] R. Yu, H. Park, and J. Lee, "Human dynamics from monocular video with dynamic camera movements," *ACM Trans. Graph.*, vol. 40, no. 6, pp. 1–14, 2021.
[223] Z. Xie, J. Tseng, S. Starke, M. van de Panne, and C. K. Liu, "Hierarchical planning and control for box loco-manipulation," *Proc. ACM Comput. Graph. Interact. Tech.*, vol. 6, no. 3, pp. 1–18, 2023.
[224] M. Hassan, Y. Guo, T. Wang, M. Black, S. Fidler, and X. B. Peng, "Synthesizing physical character-scene interactions," in *Proc. ACM SIGGRAPH Conf.*, 2023, pp. 1–9.
[225] F. Wu, Z. Gu, Y. Zhao, and A. Wu, "Learn to teach: Sample-efficient privileged learning for humanoid locomotion over real-world uneven terrain," *IEEE Robot. Automat. Lett.*, vol. 10, no. 9, pp. 9048–9055, 2025, doi: 10.1109/LRA.2025.3592131.
[226] D. Youm, H. Jung, H. Kim, J. Hwangbo, H.-W. Park, and S. Ha, "Imitating and finetuning model predictive control for robust and symmetric quadrupedal locomotion," *IEEE Robot. Autom. Lett.*, vol. 8, no. 11, pp. 7799–7806, Nov. 2023.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

[227] L. M. Smith et al., "Learning and adapting agile locomotion skills by transferring experience," in *Proc. Robot.: Sci. Syst.*, vol. 19, 2023, pp. 1–16, doi: 10.15607/RSS.2023.XIX.051.
[228] F. Jenelten, J. He, F. Farshidian, and M. Hutter, "DTC: Deep tracking control," *Sci. Robot.*, vol. 9, no. 86, 2024, Art. no. eadh5401.
[229] D. Kang, J. Cheng, M. Zamora, F. Zargarbashi, and S. Coros, "Rl model-based control: Using on-demand optimal control to learn versatile legged locomotion," *IEEE Robot. Autom. Lett.*, vol. 8, no. 10, pp. 6619–6626, Oct. 2023.
[230] S. Levine and V. Koltun, "Guided policy search," in *Proc. Int. Conf. Mach. Learn.*, vol. 28, no. 3, 2013, pp. 1–9.
[231] D. Marew, N. Perera, S. Yu, S. Roelker, and D. Kim, "A biomechanics-inspired approach to soccer kicking for humanoid robots," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2024, pp. 722–729.
[232] T. Kwon, Y. Lee, and M. Van De Panne, "Fast and flexible multilegged locomotion using learned centroidal dynamics," *ACM Trans. Graph.*, vol. 39, no. 4, pp. 46–1, 2020.
[233] A. J. Ijspeert, J. Nakanishi, and S. Schaal, "Learning attractor landscapes for learning motor primitives," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2002, pp. 1547–1554.
[234] Z. Xie, G. Berseth, P. Clary, J. Hurst, and M. van de Panne, "Feedback control for cassie with deep reinforcement learning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2018, pp. 1241–1246.
[235] R. P. singh, Z. Xie, P. Gergondet, and F. Kanehiro, "Learning bipedal walking for humanoids with current feedback," *IEEE Access*, vol. 11, pp. 82013–82023, 2023.
[236] Y. Fuchioka, Z. Xie, and M. Van de Panne, "Opt-mimic: Imitation of optimized trajectories for dynamic quadruped behaviors," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 5092–5098.
[237] F. Liu et al., "Opt2skill: Imitating dynamically-feasible whole-body trajectories for versatile humanoid loco-manipulation," *IEEE Robot. Automat. Lett.*, vol. 10, no. 11, pp. 12261–12268, 2024, doi: 10.1109/LRA.2025.3620620.
[238] C. Li, E. Stanger-Jones, S. Heim, and S. bae Kim, "FLD: Fourier latent dynamics for structured motion representation and learning," in *Proc. Int. Conf. Learn. Representations*, vol. 12, 2024, pp. 1–28. [Online]. Available: https://openreview.net/forum?id=xsd2llWYSA
[239] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learn. Representations*, vol. 2, 2014, pp. 1–14..
[240] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
[241] S. Bohez et al., "Imitate and repurpose: Learning reusable robot movement skills from human and animal behaviors," 2022, *arXiv:2203.17138*.
[242] C. Tessler, Y. Kasten, Y. Guo, S. Mannor, G. Chechik, and X. B. Peng, "Calm: Conditional adversarial latent models for directable virtual characters," in *Proc. ACM SIGGRAPH Conf.*, 2023, pp. 1–9.
[243] Z. Luo et al., "Universal humanoid motion representations for physics-based control," in *Proc. Int. Conf. Learn. Representations*, vol. 12, 2024, pp. 1–17. [Online]. Available: https://openreview.net/forum?id=OrOd8PxOO2
[244] D. Ha and J. Schmidhuber, "World models," *CoRR*, vol. abs/1803.10122, pp. 1–21, 2018. [Online]. Available: http://dblp.uni-trier.de/db/journals/corr/corr1803.html#abs-1803-10122
[245] T. Hester, M. Quinlan, and P. Stone, "Generalized model learning for reinforcement learning on a humanoid robot," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2010, pp. 2369–2374.
[246] X. Gu et al., "Advancing humanoid locomotion: Mastering challenging terrains with denoising world model learning," in *Proc. Robot.: Sci. Syst.*, vol. 20, 2024, pp. 1–14, doi: 10.15607/RSS.2024.XX.058.
[247] N. Hansen, J. S. V, V. Sobal, Y. LeCun, X. Wang, and H. Su, "Hierarchical world models as visual whole-body humanoid controllers," in *Proc. Int. Conf. Learn. Representations*, vol. 13, 2025, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=7wuJMvK639
[248] H. Shi et al., "An efficient model-based approach on learning agile motor skills without reinforcement," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 5724–5730.
[249] J. Dao, H. Duan, and A. Fern, "Sim-to-real learning for humanoid box loco-manipulation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 16930–16936.
[250] H. Munn, B. Tidd, D. Howard, and M. Gallagher, "Whole-body dynamic throwing with legged manipulators," 2024, *arXiv:2410.05681*.
[251] C. Yang, C. Pu, G. Xin, J. Zhang, and Z. Li, "Learning complex motor skills for legged robot fall recovery," *IEEE Robot. Autom. Lett.*, vol. 8, no. 7, pp. 4307–4314, Jul. 2023.
[252] M. Liu et al., "Visual whole-body control for legged loco-manipulation," in *Proc. 8th Conf. Robot Learn.*, vol. 270, 2024, pp. 1–24.
[253] Z. He, K. Lei, Y. Ze, K. Sreenath, Z. Li, and H. Xu, "Learning visual quadrupedal loco-manipulation from demonstrations," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2024, pp. 9102–9109.
[254] T. Portela, G. B. Margolis, Y. Ji, and P Agrawal, "Learning force control for legged manipulation," in *Proc. Int. Conf. Robot. Autom.*, 2024, pp. 15366–15372.
[255] F. Al-Hafez, G. Zhao, J. Peters, and D. Tateo, "LocoMuJoCo: A comprehensive imitation learning benchmark for locomotion," in *Proc. Robot Learn. Workshop*, vol. 37, 2023, pp. 1–6.
[256] B. Liu et al., "Libero: Benchmarking knowledge transfer for lifelong robot learning," *NeurIPS Datasets Benchmarks Track*, vol. 36, pp. 44776–44791, 2023.
[257] Y. Liu, B. Yang, L. Zhong, H. Wang, and L. Yi, "Mimicking-bench: A benchmark for generalizable humanoid-scene interaction learning via human mimicking," in *Proc. Conf. Comput. Vis. Pattern Recognit. Workshop Humanoid Agents*, 2024. [Online]. Available: https://arxiv.org/abs/2412.17730
[258] Z. Luo et al., "Smplolympics: Sports environments for physically simulated humanoids," in *Proc. Int. Conf. Learn. Representations*, vol. 13, 2025, pp. 1–19. [Online]. Available: https://openreview.net/forum?id=pblB72EmrM
[259] Q. Liao, B. Zhang, X. Huang, X. Huang, Z. Li, and K. Sreenath, "Berkeley Humanoid: A research platform for learning-based control," in *Proc. Int. Conf. Robot. Automat.*, 2025, pp. 2897–2904. [Online]. Available: https://api.semanticscholar.org/CorpusID:271571263
[260] S. Wilson et al., "The robotarium: Globally impactful opportunities, challenges, and lessons learned in remote-access, distributed control of multirobot systems," *IEEE Control Syst. Mag.*, vol. 40, no. 1, pp. 26–44, Feb. 2020.
[261] G. Cheng, K. Ramirez-Amaro, M. Beetz, and Y. Kuniyoshi, "Purposive learning: Robot reasoning about the meanings of human activities," *Sci. Robot.*, vol. 4, no. 26, 2019, Art. no. eaav1530.
[262] H. Qiao, Y.-X. Wu, S.-L. Zhong, P.-J. Yin, and J.-H. Chen, "Brain-inspired intelligent robotics: Theoretical analysis and systematic application," *Mach. Intell. Res.*, vol. 20, pp. 1–18, 2023.
[263] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, vol. 139, pp. 8748–8763.
[264] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 23716–23736.
[265] J. Mao et al., "Learning from massive human videos for universal humanoid pose control," in *Proc. ICRA Workshop: Hum.-Centered Robot Learn. Era Big Data Large Models*, 2025. [Online]. Available: https://openreview.net/forum?id=2pBjMDj6uJ
[266] J. Ho et al., "Imagen Video: High definition video generation with diffusion models," *Trans. Mach. Learn. Res.*, vol. 2022, 2022, pp. 1–18. [Online]. Available: https://openreview.net/forum?id=ETH1GDbSPw
[267] NVIDIA et al., "GR00T N1: An open foundation model for generalist humanoid robots," *CoRR*, vol. 2503.14734, pp. 1–36, 2025. [Online]. Available: https://arxiv.org/abs/2503.14734
[268] R.-Z. Qiu et al., "Humanoid policy human policy," in *Proc. 9th Annu. Conf. Robot Learn.*, Seoul, Korea, 2025.
[269] J. DelPreto et al., "Actionsense: A multimodal dataset and recording framework for human activities using wearable sensors in a kitchen environment," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 13800–13813.
[270] K. Grauman et al., "Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 19383–19400.
[271] K. Werling et al., "Addbiomechanics: Automating model scaling, inverse kinematics, and inverse dynamics from human motion data through sequential optimization," *PLOS One*, vol. 18, no. 11, pp. 1–25, 2023.
[272] R. Bommasani et al., "On the opportunities and risks of foundation models," *CoRR*, vol. abs/2108.07258, pp. 1–214, 2022. [Online]. Available: https://arxiv.org/abs/2108.07258
[273] OpenAI, "Gpt-4 technical report," 2023. [Online]. Available: https://openai.com/research
[274] A. Brohan et al., "RT-1: Robotics transformer for real-world control at scale," in *Proc. Robot.: Sci. Syst.*, vol. 19, 2023, pp. 1–31, doi: 10.15607/RSS.2023.XIX.025.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.

[275] B. Zitkovich et al., "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Proc. Conf. Robot Learn.*, vol. 229, 2023, pp. 2165–2183.
[276] Y. Jiang et al., "Vima: General robot manipulation with multimodal prompts," in *Proc. Int. Conf. Mach. Learn.*, vol. 202, 2023, pp. 1–48.
[277] M. J. Kim et al., "Openvla: An open-source vision-language-action model," in *Proc. Conf. Robot Learn.*, vol. 270, 2025, pp. 2679–2713.
[278] G. R. Team et al., "Gemini robotics: Bringing ai into the physical world," 2025, *arXiv:2503.20020*.
[279] Q. Bu et al., "Agibot world colosseo: A large-scale manipulation platform for scalable and intelligent embodied systems," *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2025, pp. 3549–3556. [Online]. Available: https://api.semanticscholar.org/CorpusID:276902669
[280] Y. J. Ma et al., "Eureka: Human-level reward design via coding large language models," in *Proc. Int. Conf. Learn. Representations*, vol. 12, 2024, pp. 1–45. [Online]. Available: https://openreview.net/forum?id=IEduRUO55F
[281] J. Liang et al., "Code as policies: Language model programs for embodied control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 9493–9500.
[282] brian ichter et al., "Do as I can, not as I say: Grounding language in robotic affordances," in *Proc. Annu. Conf. Robot Learn.*, vol. 205, 2022, pp. 1–34. [Online]. Available: https://openreview.net/forum?id=bdHkMjBJG_w
[283] Y. J. Ma et al., "DrEureka: Language model guided sim-to-real transfer," in *Proc. Robot.: Sci. Syst.*, vol. 20, 2024, pp. 1–28, doi: 10.15607/RSS.2024.XX.094.
[284] Y. Yao et al., "Anybipe: An end-to-end framework for training and deploying bipedal robots guided by large language models," in *IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2025, pp. 147990–14797, doi: 10.1109/IROS60139.2025.11246492.
[285] A. S. Chen et al., "Commonsense reasoning for legged robot adaptation with vision-language models," in *Proc. LangRob workshop CoRL*, 2023. [Online]. Available: https://doi.org/10.48550/arXiv.2310.06226
[286] J. Juravsky, Y. Guo, S. Fidler, and X. B. Peng, "Padl: Language-directed physics-based character control," in *Proc. SIGGRAPH Asia Conf. Papers*, 2022, pp. 1–9.
[287] J. Sun, Q. Zhang, Y. Duan, X. Jiang, C. Cheng, and R. Xu, "Prompt, plan, perform: Llm-based humanoid control via quantized imitation learning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 16236–16242.
[288] J. Wang, R. Dai, W. Wang, L. Rossini, F. Ruscelli, and N. Tsagarakis, "HYPERmotion: Learning hybrid behavior planning for autonomous loco-manipulation," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–27. [Online]. Available: https://openreview.net/forum?id=ma7McOiCZY
[289] I. Singh et al., "Progprompt: Generating situated robot task plans using large language models," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 11523–11530.
[290] W. Yu et al., "Language to rewards for robotic skill synthesis," in *Proc. Annu. Conf. Robot Learn.*, vol. 229, 2023, pp. 1–31. [Online]. Available: https://openreview.net/forum?id=SgTPdyehXMA
[291] Z. Jiang et al., "Harmon: Whole-body motion generation of humanoid robots from language descriptions," in *Proc. Annu. Conf. Robot Learn.*, vol. 270, 2024, pp. 1–20. [Online]. Available: https://openreview.net/forum?id=UUZ4Yw3lt0
[292] K. N. Kumar, I. Essa, and S. Ha, "Words into action: Learning diverse humanoid robot behaviors using language guided iterative motion refinement," *CoRR*, 2023, *arXiv:2310.06226*.
[293] Z. Xiao et al., "Unified human-scene interaction via prompted chain-of-contacts," in *Proc. Int. Conf. Learn. Representations*, vol. 12, 2024, pp. 1–30. [Online]. Available: https://openreview.net/forum?id=1vCnDyQkjg
[294] D. Totsila, Q. Rouxel, J.-B. Mouret, and S. Ivaldi, "Words2contact: Identifying support contacts from verbal instructions using foundation models," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2024, pp. 9–16.
[295] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–15.
[296] K. Black et al., "π_0: A vision-language-action flow model for general robot control," in *Proc. Robot. : Sci. Syst.*, vol. 22, 2026, pp. 1–17.
[297] A. O'Neill et al., "Open x-embodiment: Robotic learning datasets and rt-x models : Open x-embodiment collaboration0," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 6892–6903.
[298] S. Reed et al., "A generalist agent," *Trans. Mach. Learn. Res.*, pp. 1–42, 2022. [Online]. Available: https://openreview.net/forum?id=1ikK0kHjvj
[299] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. Int. Conf. Learn. Representations*, vol. 12, 2024.
[300] D. Ghosh et al., "Octo: An open-source generalist robot policy," in *Proc. Robot.: Sci. Syst.*, vol. 20, 2024, pp. 1–15, doi: 10.15607/RSS.2024.XX.090.
[301] S. Liu et al., "RDT-1b: A diffusion foundation model for bimanual manipulation," in *Proc. Int. Conf. Learn. Representations*, vol. 13, 2025, pp. 1–28. [Online]. Available: https://openreview.net/forum?id=yAzN4tz7oI
[302] C.-L. Cheang et al., "Gr-2: A generative video-language-action model with web-scale knowledge for robot manipulation," *CoRR*, vol. 2410.06158, 2024. [Online]. Available: https://doi.org/10.48550/arXiv.2410.06158
[303] H. Touvron et al., "Llama: Open and efficient foundation language models," *CoRR*, vol. 2302.13971, 2023, pp. 1–27.
[304] D. Kim, H. Woo, and Y. Lee, "Addressing bias and fairness using fair federated learning: A synthetic review," *Electronics*, vol. 13, no. 23, 2024, Art. no. 4664.
[305] P. Wensing, "Optimization and control of dynamic humanoid running and jumping," Ph.D. dissertation, Dept. Elect. Comput. Eng., The Ohio State Univ., Columbus, OH, USA, 2014.
[306] J. Carpentier and N. Mansard, "Multicontact locomotion of legged robots," *Trans. Robot.*, vol. 34, no. 6, pp. 1441–1460, 2018.
[307] S. Lengagne, J. Vaillant, E. Yoshida, and A. Kheddar, "Generation of whole-body optimal dynamic multi-contact motions," *Int. J. Robot. Res.*, vol. 32, no. 9/10, pp. 1104–1119, 2013.
[308] Z. Zhuang and H. Zhao, "Embrace Collisions: Humanoid shadowing for deployable contact-agnostics motions," in *Proc. Annu. Conf. Robot Learn.*, vol. 305, 2025, pp. 3720–3735. [Online]. Available: 3720-3735
[309] D. Kim, S. J. Jorgensen, J. Lee, J. Ahn, J. Luo, and L. Sentis, "Dynamic locomotion for passive-ankle biped robots and humanoids using whole-body locomotion control," *Int. J. Robot. Res.*, vol. 39, no. 8, pp. 936–956, 2020.
[310] S. Haddadin and E. Croft, *Physical Human–Robot Interaction.* Cham, Switzerland: Springer International Publishing, 2016, pp. 1835–1874.
[311] G. Ficht and S. Behnke, "Bipedal humanoid hardware design: A technology review," *Curr. Robot. Rep.*, vol. 2, pp. 201–210, 2021.

Zhaoyuan Gu (Student Member, IEEE) received the B.S. degree in mechanical engineering from Tsinghua University, Beijing, China, in 2018 and the M.S. degree in mechanical engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2020. He is currently working toward the Ph.D. degree in robotics with the Georgia Institute of Technology, Atlanta, GA, USA.

His research focuses on learning-based approaches for loco-manipulation of humanoid robots.

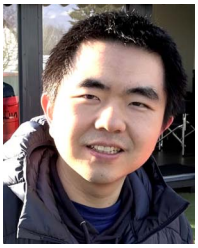
Junheng Li (Student Member, IEEE) is currently working toward the Ph.D. degree in mechanical engineering with the University of Southern California (USC), Los Angeles, CA, USA.

He has been a Research Assistant with the Dynamic Robotics and Control Laboratory (DRCL). His research centers on hierarchical optimization-based control strategies for humanoid robots to achieve dynamic and adaptive locomotion and manipulation behaviors.

Wenlan Shen (Graduate Student Member, IEEE) received the B.Eng. degree in automation engineering in 2019 from North China Electric Power University, Beijing, China, and the M.Sc. degree in electrical and computer engineering in 2023 from the Technical University of Munich, Munich, Germany, where she is currently working toward the Ph.D. degree in robotics with the Institute for Cognitive Systems.

Her current research interests include whole-body manipulation, multirobot collaboration, and human–robot interaction.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.



Wenhao Yu received the bachelor's degree in software engineering from Shanghai Jiao Tong University, Shanghai, China, in 2013, and the Ph.D. degree in computer science from the Georgia Institute of Technology, Atlanta, GA, USA, in 2020, advised by Karen Liu and Greg Turk.

He is currently a Staff Research Scientist with Google DeepMind Robotics team. His research aims to build intelligent, safe, and helpful robots that can operate in diverse environments by harnessing tools from reinforcement learning, foundation models, and control.

Zhaoming Xie (Member, IEEE) received the Ph.D. degree in computer science from the University of British Columbia, Vancouver, BC, Canada, in 2021.

He is currently a Research Scientist with the Robotics AI Institute. He spent a year as a postdoc with Stanford University, Stanford, CA, USA. His research interest is in the control of legged robots, with tools from machine learning and optimal control.

Stephen McCrory is currently working toward the Ph.D. degree with the Intelligent Systems and Robotics Program, University of West Florida, Pensacola, FL, USA.

He is currently a Senior Research Associate with the Florida Institute for Human and Machine Cognition (IHMC), Pensacola, FL, USA. His research focuses on deploying multicontact control and teleoperation approaches to IHMC's humanoid robot Nadia.

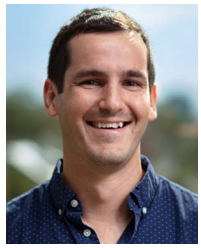
Xianyi Cheng (Member, IEEE) received the Ph.D. degree in mechanical engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2024.

She is currently an Assistant Professor with the Department of Mechanical Engineering and Materials Science, Duke University, Durham, NC, USA, where she directs the Robot Dexterity Lab. Her research focuses on robotic dexterity and manipulation.

Dr. Cheng was the recipient of the Foxconn Graduate Fellowship in 2018 and was selected for the 2021 MIT EECS Rising Stars.

Abdulaziz Shamsah (Member, IEEE) received the B.S. degree in mechanical engineering from the Rensselaer Polytechnic Institute, Troy, NY, USA, in 2016 and the M.S.E. degree in mechanical engineering and applied mechanics from the University of Pennsylvania, Philadelphia, PA, USA, in 2018, and the Ph.D. degree in mechanical engineering from the Georgia Institute of Technology, Atlanta, GA, USA, in 2024.

He is currently an Assistant Professor with the Mechanical Engineering Department, College of Engineering and Petroleum, Kuwait University, Kuwait City, Kuwait. His research interests include bipedal navigation in real-world environments, formal methods, and safety and robustness in locomotion.



Robert Griffin (Member, IEEE) received the bachelor' degree in mechanical engineering from Tennessee Technological University, Cookeville, TN, USA, in 2012, and the Ph.D. degree in mechanical engineering from Virginia Tech, Blacksburg, VA, USA, in 2017.

He leads the Robotics Group, Florida Institute for Human and Machine Cognition (IHMC), Pensacola, FL, USA, heading a number of projects focusing on advancing humanoid robots. His research interests include system level approaches for improving the mobility and capability of these robotic platforms, including platform design, motion design and control, autonomy and manipulation, and perception. These efforts are mostly centered around IHMC's humanoid Nadia, designed to have a high power-to-weight ratio and large range of motion.

C. Karen Liu received the bachelor' degree in computer science from National Taiwan University, Taipei City, Taiwan, in 1999, the M.S. and Ph.D. degrees in computer science from University of Washington, Seattle, WA, USA, in 2001 and 2005, respectively.

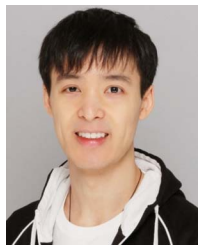
She is currently a Professor with the Computer Science Department, Stanford University, Stanford, CA, USA. Her research interests include computer graphics and robotics, including physics-based animation, optimal control, and reinforcement learning. She developed computational approaches to modeling realistic and natural human movements, learning complex control policies for humanoids and assistive robots, and advancing fundamental numerical simulation and optimal control algorithms.

Ms. Liu was the recipient of a National Science Foundation CAREER Award, an Alfred P. Sloan Fellowship, and was named Young Innovators Under 35 by Technology Review.

Abderrahmane Kheddar (Fellow, IEEE) received the bachelore's degree in computer science from École Nationale Supérieure d'Informatique, Oued Smar, Algeria, in 1990, and the M.Sc. and Ph.D. degrees in virtual reality from Pierre and Marie Curie University, Paris, France, in 1993 and 1997, respectively.

He is currently a Professor with CNRS. His research interests include humanoids, haptics and recently bionics.

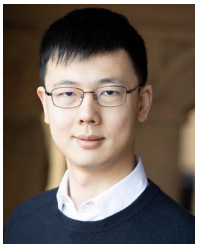
Dr. Kheddar is a founding member of the IEEE/RAS chapter on haptics, the co-chair and founding member of the IEEE/RAS Technical committee on model-based optimization, he is a member of the steering committee of the IEEE Brain Initiative, Editor of the IEEE ROBOTICS AND AUTOMATION LETTERS, Founding member and Deputy Editor-in-Chief of Cyborg and Bionics System. He was Editor of the IEEE TRANSACTIONS ON ROBOTICS (2013–2018), a founding member of the IEEE TRANSACTIONS ON HAPTICS and was an editorial board during three years (2007–2010). He is an AAIA fellow, member of the National Academy of Technology of France and knight of the national order of merits of France.



Xue Bin Peng received the bachelor's and M.Sc. degrees in computer science from University of British Columbia, Vancouver, Canada, in 2015 and 2017, respectively, and the Ph.D. degree in computer science from the University of California, Berkeley, CA, USA, in 2021

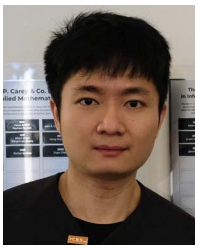
He is currently an Assistant Professor with Simon Fraser University, Burnaby, CA, USA, and a Research Scientist with NVIDIA, Santa Clara, CA, USA.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.



Yuke Zhu (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees in computer science from Stanford University, Stanford, CA, USA, in 2015 and 2019, respectively.

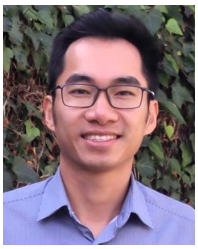
He is currently an Assistant Professor with the Department of Computer Science, University of Texas at Austin, Austin, TX, USA, and the Director of the Robot Perception and Learning (RPL) Lab. He focuses on developing intelligent algorithms for generalist robots to reason about and interact with the real world. His research spans robotics, computer vision, and machine learning.



Guanya Shi (Member, IEEE) received the Ph.D. degree in control and dynamical systems from California Institute of Technology, Pasadena, CA, USA, in 2022.

He is currently an Assistant Professor with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA, leading the Learning and Control for Agile Robotics (LeCAR) Lab. Before joining CMU, he was a Postdoctoral Scholar with the University of Washington, Seattle, WA, USA. He is broadly interested in the intersection of machine learning and control, spanning the entire spectrum from theory and foundation, algorithm design, to real-world agile robotics.

Dr. Shi was the recipient of the Outstanding Student Paper Award Finalist at RSS 2024 and the Best Paper Award Finalist at ICRA 2025. His research has been featured by major media outlets such as CNN, Reuters, and IEEE Spectrum. He is currently an Associate Editor for IEEE ROBOTICS AND AUTOMATION LETTERS.



Quan Nguyen (Member, IEEE) received the Ph.D. degree in mechanical engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2017.

He is currently an Assistant Professor of Aerospace and Mechanical Engineering with the University of Southern California (USC), Los Angeles, CA, USA. Prior to joining USC, he was a Postdoctoral Associate with the Biomimetic Robotics Lab, Massachusetts Institute of Technology, Cambridge, MA, USA. His research interests span different control and optimization approaches for highly dynamic robotics including nonlinear control, trajectory optimization, real-time optimization-based control, robust and adaptive control. His work on the bipedal robot ATRIAS walking on stepping stones was featured on the IEEE Spectrum, TechCrunch, TechXplore and Digital Trends.

Dr. Nguyen was the recipient of the Best Dissertation Award for his Ph.D. degree. His work on the MIT Cheetah 3 robot leaping on a desk was featured widely in many major media channels, including CNN, BBC, NBC, ABC, etc. He was also a recipient of the Best Presentation of the Session at the 2016 American Control Conference and the Best System Paper Finalist at the 2017 Robotics: Science Systems.

Gordon Cheng (Fellow, IEEE) received the B.Sc. and M.Sc. degrees in computer science from the University of Wollongong, Wollongong, NSW, Australia, in 1991 and 1993, respectively, and the Ph.D. degree in systems engineering from the Australian National University, Canberra, ACT, Australia, in 2001.

He has been a Chair Professor and Director of the Institute for Cognitive Systems, Technical University of Munich, Munich, Germany, since 2010, Coordinator of the Center of Competence Neuro-Engineering since 2013, and the Director of the Elite Master of Science Program in Neuroengineering (MSNE) of the Elite Network of Bavaria since 2016. He is also the coinventor of approximately 20 patents and has coauthored more than 450 technical publications, proceedings, editorials, books, and book chapters. His current research interests include humanoid robotics, neurorobotics, imitation learning, cognitive systems, artificial intelligence, and neuroengineering.

Dr. Cheng was named IEEE Fellow in 2017 for his contributions to humanoid robotic systems and neurorobotics, and was elected Fellow of the Robotics Society of Japan in 2024 for his contributions to the advancement of robotics research.

Huijun Gao (Fellow, IEEE) received the Ph.D. degree in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 2005.

From 2005 to 2007, he was a Postdoctoral Researcher with the Department of Electrical and Computer Engineering, University of Alberta, Edmonton, AB, Canada. Since 2004, he has been with the Harbin Institute of Technology, where he is currently a Chair Professor and the Director of the Research Institute of Intelligent Control and Systems. His research interests include intelligent and robust control, robotics, mechatronics, and their engineering applications.

Dr. Gao is a member of Academia Europaea and Vice President of the IEEE Industrial Electronics Society. He is/was the Editor-in-Chief of IEEE/ASME TRANSACTIONS ON MECHATRONICS, Co-Editor-in-Chief of IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, and Associate Editor for *Automatica*, IEEE TRANSACTIONS ON CYBERNETICS, and IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS. He was the recipient of the 2022 Dr.-Ing. Eugene Mittelmann Achievement Award and the 2023 Norbert Wiener Award and is a Distinguished Lecturer of the IEEE Systems, Man and Cybernetics Society.

Ye Zhao (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from The University of Texas at Austin, Austin, TX, USA, in 2016.

He was a Postdoctoral Fellow with the John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA. He is currently an Associate Professor with George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, GA, USA. His research interests include robust task and motion planning, contact-rich trajectory optimization, and formal methods for legged locomotion and navigation.

Dr. Zhao was the recipient of the CAREER Award from the National Science Foundation in 2022 and the Young Investigator Award from the Office of Naval Research in 2023. He is currently an Associate Editor for IEEE TRANSACTIONS ON ROBOTICS, IEEE/ASME TRANSACTIONS ON MECHATRONICS, IEEE ROBOTICS AND AUTOMATION LETTERS, and IEEE CONTROL SYSTEMS LETTERS.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 22,2026 at 06:20:32 UTC from IEEE Xplore. Restrictions apply.